MILITARY SPECIFICATION

# RADIO SET AN/GRC-103( )(V) (BASIC)

## 1. SCOPE

1.1 This specification covers the requirement for the construction of a type radio relay equipment for use in any of several bands in the frequency range 76 to 1000 MHZ. and 1350 to 1850 MHZ.

1.2 Radio Set AN/GRC-103( )(V) consists of the following components.

1.2.1 Common Units. Common units are required for operation in any of the several frequency ranges.

1 Receiver, Radio R-1329(P)( )/GRC-103(V) includes 1 ea case, Receiver, CY-4638( )/GRC-103(V).

1 Transmitter Radio T-983(P)( )/GRC-103(V) includes 1 ea case,CY-4637( )/GRC-103(V).

1 Receiver - Transmitter, Orderwire RT-773( )/GRC-103(V) includes 1 ea case, Receiver - Transmitter Orderwire CY-4635( )/GRC-103(V).

1 Mast, AB-952( )/GRC-103(V) includes 1 ea Elevator, Antenna AB-1072( )/GRC-103(V), 7 ea Mast Section AB-1071( )/GRC-103(V) and 1 ea Accessory Kit, MK-1069( ) /GRC-103(V).

2 Cable Assembly, RF CG-3444( )/U (1 ft 6 in).

1 Cable Assembly, RF-CG-3444( )/U (3 ft 6 in).

1 Cable Assembly, RF CG-3443( )/U (80 ft).

1 Adaptor, Connector UG-1375( )/U.

2 Cable Assembly, Special Purpose, Electrical CX-10762( )/U (5 ft).

1 Cable Assembly, Special Purpose, Electrical, Branched CX-10763( ) /GRC-103(V) (3 ft 6 in).

1 Dummy Load, Electrical DA-437( )/GRC-103(V).



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1.2.2 Frequency band units. Frequency band units are those components peculiar to one operating band as specified in the bid request and contract.

1.2.2.1 Band zero components (76-170 MHZ)

1 ea Amplifier - Converter AM-6051( )/GRC-103(V).

1 ea Amplifier - Frequency Multiplexer AM-6052( )/GRC-103(V).

1 ea Antenna AS-2235( )/GRC-103(V).

1.2.2.2 Band I components (220-405 MHZ)

1 ea Amplifier Converter AM-4316( )/GRC-103(V).

1 ea Amplifier Frequency Multiplier AM-4320( )/GRC-103(V).

1 ea Antenna AS-1852( )/GRC-103(V), includes 1 ea Reflector, Antenna AS-2150( )/GRC-103(V) and 1 ea Antenna Element AS-2151( )/GRC-103(V).

1.2.2.3 Band II Components (395 to 705 MHZ)

1 ea Amplifier Converter AM-4317( )/GRC-103(V).

1 ea Amplifier Frequency Multiplier AM-4321( )/GRC-103(V).

1 ea Antenna AS-1853( )/GRC-103(V), consisting of 1 ea Reflector, Antenna AS-2150( )/GRC-103(V), and 1 ea Antenna Element AS-2194( )/GRC-103(V).

1.2.2.4 Band III Components (695 to 1000 MHZ)

1 ea Amplifier Converter AM-4318( )/GRC-103(V).

1 ea Amplifier Frequency Multiplier AM-4322( )/GRC-103(V).

1 ea Antenna AS-1854( )/GRC-103(V), consisting of 1 ea Reflector, Antenna AS-2150( )/GRC-103(V) and 1 ea Antenna Element AS-2195( )/GRC-103(V).



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1.2.2.5 Band IV components (1350-1850 MHZ)

1 ea Amplifier Converter AM-4319( )/GRC-103(V).

1 ea Amplifier, Frequency Multiplier AM-4322( )/ GRC-103(V).

1 ea Antenna AS- ( )/GRC-103(V).

1.2.3 Special accessories. Special accessories are used with the basic radio set for extension of capabilities and/or for maintenance.

1 ea Extension Kit, Mast MK-1009( )/GRC-103(V) including case, Mast Extension Kit CY-6148( )/GRC-103(V).

1 ea Converter, Frequency Electronic CV-2500( )/GRC.

1 ea Cable Assembly, Special Purpose Electrical CX-10879( )/GRC-103(V).

1 ea Amplifier, Audio Frequency AM-4324( )/GRC-103(V) includes Case, Amplifier, Audio Frequency CY-4636( )/ GRC-103(V).



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## 2. APPLICABLE DOCUMENTS

The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

### SPECIFICATIONS

#### FEDERAL

| | |
|---|---|
| T-T-871 | Twine, cotton, wrapping |
| QQ-W-343 | Wire, Electrical (uninsulated) |
| PPP-B-636 | Box, Fiberboard |
| PPP-B-640 | Box, Fiberboard, corrugated, triple-wall |
| PPP-F-320 | Fiberboard, corrugated and solid, sheet stock (container grade) and cut shapes |
| PPP-P-291 | Paperboard, wrapping, cushioning |
| PPP-F-760 | Strapping, Nonmetallic, (and connectors) |
| PPP-T-76 | Tape, pressure-sensitive adhesive paper, water resistant,(for carton sealing) |
| PPP-T-97 | Tape, Pressure-sensitive adhesive, filament reinforced. |

#### MILITARY

| | |
|---|---|
| MIL-C-17 | Cables, Radio Frequency, Coaxial, Dual Coaxial, twin conductor, and twin lead. |
| MIL-P-116 | Preservation, methods of |
| MIL-T-152 | Treatment, moisture-and-fungus-resistant, of communications, electronic, and associated electrical equipment. |
| MIL-V-173 | Varnish, moisture-and-fungus-resistant (for the treatment of communication, electronic, and associated electrical equipment). |
| MIL-S-901 | Shock tests, H.I. (High-Impact), Shipboard machinery, equipment and systems, requirement for (NAVY). |



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| | |
|---|---|
| MIL-C-3432 | Cable and wire, Electrical (Power and control, Flexible and extra flexible, 300 and 600 volts). |
| MIL-C-3702 | Cable, Power, Electrical, Ignition, High Tension |
| MIL-C-3849 | Cord, Electrical (tinsel). |
| MIL-C-3883 | Cord, Electrical (Audio Frequency) |
| MIL-C-3884 | Cord, Electrical (Short Lay) |
| MIL-C-3885 | Cable Assemblies and cord assemblies, electrical (for use in electronic, communication and associated electrical equipment) |
| MIL-I-3930 | Insulating and jacketing compounds, electrical (for cables, cords, and wires) |
| MIL-C-11060 | Cables, twisted pair internal hookup, unshielded |
| MIL-P-11268 | Parts, materials and processes used in electronic communication equipment |
| MIL-M-11473 | Magnetic-Particle Inspection, soundness requirements for weldments. |
| MIL-H-11996 | Handset H-60( )/PT, Handset H-156( )/U and Handset H-165( )/U |
| MIL-M-13231 | Marking of Electronic Items |
| MIL-C-13777 | Cable, Special Purpose, Electrical, General Specification for |
| MIL-F-14072 | Finishes for Ground Signal Equipment |
| MIL-E-15090 | Enamel, equipment, light-gray (Formula No. 111) |
| MIL-C-22931 | Cable, Radio Frequency, semirigid, coaxial, semi-air-dielectric, general specification for |
| MIL-B-43291 | Boxes, fiberboard, corrugated, double wall, weather resistant |

NOTE: See Appendix I for list of Test Specifications

STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-595 | Colors |

MILITARY

| | |
|---|---|
| MIL-STD-16 | Electrical and Electronic Reference Designations |
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |



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| | |
|---|---|
| MIL-STD-108 | Definitions of and Basic Requirements for Electrical and Electronic Equipment |
| MIL-STD-252 | Wired Equipment, Classification of Visual and Mechanical Defects |
| MIL-STD-454 | Standard General Requirements for Electronic Equipment |
| MIL-STD-461 | Electromagnetic Interference Characteristics, Requirements for Equipment |
| MIL-STD-462 | Electromagnetic Interference Characteristics, Measurement of |
| MIL-STD-681 | Identification Coding and Application of Hookup and Lead Wire |
| MIL-STD-781 | Test Levels and Accept/Reject Criteria for Reliability of Non-Expendable Electronic Equipment |
| MIL-STD-810 | Environmental Test Methods |

USAECOM

| | |
|---|---|
| RDD-STD-1 | Microelectronics (Integrated Electronics) and Modular Design in Electronic Equipment |

DRAWINGS

ELECTRONICS COMMAND

| | |
|---|---|
| SC-A-46439 | List of Accessories for Package Tester |

PUBLICATIONS

DEPARTMENT OF THE ARMY

| | |
|---|---|
| TM-11-5820-540-12 | Operator and Organizational Maintenance Manual-Radio Set AN/GRC-103(V) |
| TM-11-5820-540-35 | DS, GS, and Depot Maintenance Manual-Radio Set AN/GRC-103(V) |

DEFENSE SUPPLY AGENCY

| | |
|---|---|
| MIL-HDBK-216 | RF Transmission Lines and Fittings (Application for copies should be addressed to the Commanding Officer, US Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120) |



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# 3. REQUIREMENTS

## 3.1 Description

3.1.1 Functional. The equipments covered by this specification are units of a complete, compact, rugged, radio relay equipment whose major operating components may be used either as a terminal or modulation frequency patch repeater for intermediate or forward area systems. The radio set is used to provide simultaneous two-way (duplex) transmission. The system provides radio-frequency (rf) channel assignments in the 76 MHZ. to 1850 MHZ. When used in combination, either as a repeater or as a terminal, it is possible to tune and operate the receivers and transmitters on independent frequencies over the frequency range. The basic radio set contains a maximum of common equipment with the plug-in units containing only those circuits which are peculiar to each frequency range.

3.1.2 Radio Set AN/GRC-103( )(V)(1) physical. The (V)1 version provides for multichannel, line-of-sight, two-way communication in the VHF - UHF frequency range. It provides 370 operating channels in the 220-404.5 MHZ range (Channels 40-409). This version is intended primarily for use as a radio link in a communication network, which may include carrier telephone and teletypewriter sets. The radio set is used at each end of the line-of-sight path. The radio set may be used with 4 channel frequency division multiplex (fdm), or 6/12 or 24 channel pulse code modulation (pcm) equipment. The following major components comprise Radio Set AN/GRC-103( )(V).

3.1.2.1 Transmitter Radio T-983(P)( )/GRC-103(V). The transmitter for Radio Set AN/GRC-103( )(V)( ) consists of a Modulator-Power Supply Group and Case, Transmitter CY-4637/GRC-103(V). The case provides protection for the Modulator-Power Supply Group and the Amplifier-Frequency Multiplier AM-4320/GRC-103(V). Integral to the case are interconnections for the plug-in units and a built-in blower to provide cooling for the power amplifier tubes. The modulator-power supply group contains those modules common to all bands of operation of the basic radio set.

3.1.2.2 Amplifier-Frequency Multiplier AM-4320( )/GRC-103(V). The amplifier-frequency multiplier contains those modules unique to operation of the transmitter of the radio set in the range 220 to 404.5 MHZ. It provides the proper frequency multiplication factors and frequency selection necessary for operation in Band I.



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3.1.2.3 Receiver, Radio R-1329(P)( )/GRC-103(V). The receiver for Radio Set AN/GRC-103(V) consists of a radio receiver group and case, Receiver CY-4632( )/GRC-103(V). The case provides protection for the radio receiver groups and Amplifier-Converter AM-4316( )/GRC-103(V) and contains integral wiring for interconnection of the two units. The radio receiver group contains those modules common to all bands of operation of the basic radio set.

3.1.2.4 Amplifier-Converter AM-4316( )/GRC-103(V). The amplifier converter contains those modules unique to operation of the receiver of the radio set in the ranges 220-404.5 only. It provides the proper frequency multiplication factors and frequency selection necessary for operation in Band I.

3.1.2.5 Receiver-Transmitter Order Wire RT-773( )/GRC-103(V). The order wire unit provides an engineering channel, separate from traffic channels, for alignment and adjustment of the system.

3.1.2.6 Dummy Load, Electrical DA-437( )/GRC-103(V). The dummy load provides a facility for tuning the radio set and testing without the use of an antenna.

3.1.2.7 Antenna AS-1852( )/GRC-103(V). The antenna provides for transmission and reception of radio frequency signals between radio sets in the frequency range 220 to 404.5 MHZ.

3.1.2.8 Mast AB-952( )/GRC-103(V). The mast provides a thirty-five foot means of support for the antenna.

3.1.2.9 Extension Kit Mast MK 1009( )/GRC-103(V). This kit when used with Mast AB-952( )/GRC-103(V) extends the height of the mast to fifty feet.

3.1.2.10 Cables (operating). Cables to permit operation of the Radio Set AN/GRC-103( )(V)(1) shall be provided as specified in the bid request and contract.

a. Cable, rf, CG-3444/U (1-1/2') two each, to connect transmitter to duplexer, and Adapter UG-1375/U to antenna.

b. Cable, rf, CG-3444/U (3-1/2') to connect duplexer to Adapter UG-1375/U.

c. Cable, rf, CG-3443/U (80') for a transmission line between the Radio Set AN/GRC-103(V)1 and the antenna.



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d. Cable, Special Purpose Electrical, Branched CX-10763/U to interconnect order wire to receiver and transmitter.

e. Cable, Electrical, Special Purpose CX-10762/U (5')

3.1.2.11 Connector UG-1375/U. To connect Cable, RF CG-3444/U (3-1/2') to Cable, RF CG-3443/U (80').

3.2 Preproduction samples. The contractor shall furnish a lot of thirteen preproduction samples for approval, or as otherwise specified in the bid request and contract. The equipment shall be tested as follows:

a. Complete conformance testing, 3 each.

b. Reliability testing, ten each.

3.3 Procurement model. A procurement model of the equipment will be available for inspection by prospective bidders and will be lent to the contractor. Unless otherwise specified herein or in the invitation for bids, the equipment shall conform to the following:

3.3.1 Physical Construction. Physical construction of the equipment shall conform to the procurement model except for parts, materials and processes which shall conform to the requirements of 3.4. Where requirements of this specification and its subsidiary specifications dictate dimensional changes or effect interchangeability of assemblies, subassemblies or modules with the model, the contracting officer shall be notified in writing and changes shall be made only after prior approval of the contracting officer.

3.3.2 Features. The equipment shall incorporate all circuit functions and operational features of the procurement model. All components shall conform to 3.4.

3.3.3 Performance. The contractor shall adjust and [illegible] the procurement model for optimum performance and then shall measure its performance characteristics. Performance of the equipment shall conform to those measurements, except that, in case of conflict between specified performance characteristics and the performance of the procurement model, specified performance characteristics shall govern.



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3.4 Parts, materials and procresses. In addition to any requirements of this specification covering specific parts, materials and processes, all electronic components, parts devices, materials, techniques and technologies shall conform with MIL-P-11268 except for the following:

a. Microelectronics (integrated electronics) and modular design shall be in accordance with RDD-STD-1.

b. Plating and coating. Conductor patterns shall be in accordance with 5.1.3 and 5.1.4 of MIL-STD-275. Metallic coating, if specified, shall be limited to one of the following: (1) Solder coated (a minimum of 0.0003 inch thick conforming to composition SN-60 or SN 63 of QQ-S-571); or (2) Gold plated (0.000050 to 0.000080 inch thick). Such metallic coating or plating of conductors is permissible but not required. Conductor dimensions shall be of adequate size to provide sufficient conductivity so that the maximum safe operating temperature of the base material will not be exceeded during normal operations (reference Fig. 1 of MIL-STD-275). Conductor line width shall be 0.015 inch minimum.

c. One-part PC connector. Since the one-part printed circuit connector utilizes exposed conductor tabs and cannot be coated, it is not recommended for general use in army equipment. Under special circumstances of assured low humidity (less than 50% HR) applications (e.g., air conditioned fixed plant installation) and where creepage distances of 0.080" minimum and voltage difference of 50V (DC or peak AC) maximum can be maintained, one-part connectors may be used provided the receptacle meets the requirements of MIL-C-21097/1, "Connector, Electrical, Receptacle, Card Insertion, Printed Wiring, Contact Spacing (0.156) Type A and AD", and the printed wiring plug portion meets the requirements of ES-C-199581, "Requirements for Printed Card Edge Connectors Mating with MIL-C-21097/1 Connector Receptacles".

3.5 Finish, protective. Equipment shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware type items, such as handles, hinges, screws, etc., and necessary touch-up after mounting. Unless otherwise specified, the final paint film on type I surface (see MIL-F-14072), exposed to view during operation, shall be a smooth, semigloss (olive drab) green, in color conforming to No. X24087 of FED-STD-595. When some other color or gloss is referred to in the equipment specification, but the details are not specified, the contractor shall apply to the



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contracting officer for information regarding the required finish. For interference reduction purposes, interface surfaces of structures and bonding connections between cases and mounting bases shall have a finish for electrical contact. The top coat of paint on the front panel of the equipment shall be semi-gloss gray enamel in accordance with MIL-E-15090.

3.6 Marking.

3.6.1 General. Marking shall conform to MIL-M-13231. Front panel marking shall be group I as described in that specification.

3.6.2 Visibility. Wherever practicable, parts shall be so mounted that their identification marking will be readily visible with minimum disassembly of the equipment.

3.6.3 Chassis marking. All chassis identification, parts identification, warning notes, etc., shall be applied by the silk screen process or processes giving the same degree of legibility, placement accuracy and permanence. The marking shall be in accordance with the model. Assemblies, subassemblies and modules serially numbered in the model shall have serial numbers in addition to the major units specified in 3.6.4.

3.6.4 Serial numbers. The following items shall have serial numbers and name plates.

Transmitter, Radio T-983( )P/GRC-103(V)

Amplifier-Frequency Multiplier AM-4320( )/GRC-103(V)

Receiver, Radio R-1329 ( )P/GRC-103(V)

Amplifier-Converter AM-4316( )P/GRC-103(V)

Receiver-Transmitter, Order Wire RT-773( )/GRC-103(V)

Antenna AS-1852( )/GRC-103(V)

Mast AB-952( )/GRC-103(V)

Dummy Load, Electrical DA-437( )/GRC-103(V)



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3.6.4.1 Composition of serial numbers. The serial numbers used for this equipment shall be ten digits. The first five digits shall be the Federal Identification number of the prime manufacturer. The sixth digit shall be a code letter to indicate the production contract with said manufacturer. A for first production, B for second production etc. The final four digits shall be progressively from 0000 to 9999 as required. For substantially continuous production on more than one contract consecutive numbering without change of letter designator may be used upon application to and approval by the Contracting Officer. Records will be maintained and furnished the Government of the first and last serial number of each unit, assembly, subassembly, and module furnished on the contract.

3.7 Tropicalization.

3.7.1 Fungus-inert materials. No fungi-susceptible materials, treated or otherwise, shall be used except in an hermetically sealed enclosure. Fungus-inert materials shall be in accordance with requirement 4 of MIL-STD-454.

3.7.2 Overall treatment (moisture and fungus proofing treatment (MFP). When overall tropicalization treatment of equipment is a contractual requirement, a moisture and fungus-resistant coating of varnish conforming to MIL-V-173 shall be applied in accordance with MIL-T-152. The moisture, fungus-resistant coating shall be applied after the equipment has been cleaned to remove such contaminants as corrosion products, dust, fingerprints, lubricating oils, mold release agents, sand, and solder fluxes. When tape is used on parts of equipment that are masked against treatments, the tape shall be an electrical grade. In the application of the MFP coating, care shall be taken to assure that the coating does not affect circuit constants and thereby degrade performance.

3.8 Identification of wiring. Wiring shall be identified by color-coding or, if approved by the contracting officer, may be identified by other means such as marking of terminals at both ends of each lead or marking on the insulation. Color coding for chassis wiring shall be system I conforming to MIL-STD-681.

3.9 Controls. Direction of rotation of turning controls shall be identical to that of the procurement model.



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3.10 Adjustment and repair. The equipment shall be so constructed that parts, terminals, wiring, etc., are accessible for circuit checking, adjustment, maintenance, repair, and replacement with minimum disturbance to other parts and wiring and with use of the minimum number of variety of special tools, particularly those needed for adjustment. When tools for adjustment are needed, the component where they are used shall provide convenient means for mounting them. Restoration of interrupted service shall be within one hour for 98% of the probable causes of malfunctions.

3.11 Electrical clearance and creepage distances. Minimum electrical clearances and creepage distances shall conform to Table I.

TABLE I

Electrical clearance and creepage distances (in inches)
(Except for printed circuits)

| Voltage, ac or dc | Clearance | Creepage |
|---|---|---|
| To 150 | 1/8 | 1/8 |
| 150-300 | 1/8 | 1/8 |
| 300-600 | 1/8 | 1/4 |
| 600-1000 | 1/4 | 3/4 |

3.12 Interchangeability. Like units, assemblies, subassemblies and repalceable parts shall conform to Requirement 7 of MIL-STD-454 and shall be physically and functionally interchangeable, without modification or adjustment except operator adjustments, of such items or of the equipment. When dimensions, ratings, characteristics, etc., are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing. If the contractor is in doubt as to whether a particular subassembly or part is to be considered replaceable, the contracting officer shall be consulted.

3.13. Electrical Requirements.

3.13.1 Functional characteristics. The equipment covered by this specification shall be a complete, compact, rugged,



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lightweight radio relay equipment whose major operating components may be used either as a terminal or modulation frequency patch repeater for primary use in the division area communications system. The radio shall provide simultaneous two way (duplex) transmission with the system performance as specified herein.

3.13.2 Frequency range. The radio set shall operate in five bands as follows: Band 0: 76 - 170 MHz, Band I: 220 - 404.5 MHz, Band II: 395 - 705 MHz, Band III: 695 -1000 MHz, and Band IV: 1350 - 1850 MHz. The basic radio set contains a maximum of common equipment with the individual band units containing those circuits which are peculiar to that band. The equipment covered by this specification shall conform to Band I requirements.

3.13.3 RF Channels and spacing. The system shall provide channel assignments on each half and full megacycles points for Bands I, II and III (1/2 MHz spacing) and on each half megacycle point for Bands IV (1 MHz spacing). Band 0 channel assignments are on 1/8 MHz points (1/8 MHz spacing).

3.13.4 RF transmitter.

3.13.4.1 Power Output. The normal transmitter power output without the rf duplexer (described in 3.13.37) shall be at least 40 watts in Band 0, 30 watts in Band I, 15 watts in Bands II, III and IV.

3.13.4.2 Power dissipation. The transmitter shall have provision for maintaining the high-level power tubes within their recommended plate dissipation ratings under condition of no excitation, or when the rf output terminals are open circuited or shorted.

3.13.4.3 Frequency stability. The transmitter when set to any channel in any band, after 15 minute warmup and at any ambient temperature between -35 degrees F and +125 degrees F, shall not vary beyond plus or minus 0.002% of the nominal channel frequency for the next eight hours. The ambient may vary up to 54 degrees F within the prescribed limits during the eight hour period

3.13.4.4 Frequency setability. It shall be possible to align the transmitter to within .002% of the desired frequency by the use of an integral indicator, calibrated in rf channel numbers.



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3.13.4.5 Spurious outputs. When measured with duplexer, the power output of any harmonic shall be at least 80 db below that of the fundamental and the power output of any non-harmonic extraneous emission of a band outside plus and minus 15 MHz of the carrier shall be at least 80 db below that of the fundamental. All spurious outputs over the frequency range $f_o \pm 15$ MHz shall be suppressed at least 60 db, except for the range occupied by the modulation, where the spurious outputs shall not exceed the level of the modulation envelope.

3.13.4.6 Alarm. A visual and audible alarm shall be provided in the transmitter. The visual alarm shall indicate any change in rf power output that is greater than 8 db $\pm$ 2 db below the nominal power output rating of the transmitter. A switch associated with the audible alarm is included such that: In one switch position the alarm shall sound when the visual alarms are lighted as in the case of a failure; in the second switch position, the alarm sounds when the light is not lighted. The audible alarm thus indicates a change in state of the visual signal. A push to operate switch shall be employed.

3.13.4.7 Transmitter frequency response. The overall frequency response of the transmitter when a modulation signal to produce 100 KC peak deviation is applied to the video input shall conform to Table II.



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TABLE II

| FREQUENCY KHZ | REL. ATTEN. dB. REFERENCE 10 KHZ | | |
|---|---|---|---|
| | NOMINAL | MINIMUM | MAXIMUM |
| .125 | 1.0 | 0.5 | 1.5 |
| .200 | .5 | 0 | 1.0 |
| .250 | .2 | 0 | .4 |
| 1.0 | 0 | -0.1 | 0.1 |
| 10 | 0 | --- | --- |
| 30 | 0 | -.2 | .2 |
| 60 | .1 | 0 | .3 |
| 120 | .15 | 0 | .5 |
| 240 | .7 | .5 | 1.3 |
| 360 | 1.3 | 1.1 | 2.3 |
| 480 | 2.7 | 2.4 | 4.4 |
| 600 | 4.2 | 3.9 | 6.9 |
| 720 | 6.0 | 5.8 | 9.8 |
| 840 | 8.2 | 8.0 | 13.0 |
| 960 | 10.7 | 9.0 | 15.0 |

3.13.4.8 Dummy antenna. The dummy antenna shall be capable of dissipating the maximum rf power delivered by the transmitter, presenting a non-reactive 50 ohm load over the entire frequency range. The VSWR shall not exceed 1.5.

3.13.5 Receiver. The following requirements shall be met with the duplexer described in 3.13.37.

3.13.5.1 Noise Figure. The noise figure of the receiver shall not exceed 8 DB between 76 to 170 MHZ (Band 0), 8 DB between 220 MHZ to 400 MHZ and 9 DB between 400 to 404.5 MHZ (Band I), 8 DB between 395 to 1000 mc (Band II and III), 8 DB between 1350 MHZ to 1850 MHZ (Band IV). These values shall include the losses due to the receiver protection device and the duplexer.

3.13.5.2 Selectivity. The selectivity of the receiver shall be such that the requirements of 3.13.5.3 and 3.13.9 are met.



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3.13.5.3 Spurious response. The ratio of described signal response to any spurious response shall be as follows:

a. For a maximum of 6 discrete responses the ratio shall be at least 80 db.

b. For all other responses the ratio shall be at least 100 db.

3.13.5.4 Receiver setability. It shall be possible to align the radio receiver to within 0.002% ±20 KC of the channel frequency by the use of an integral indicator, calibrated in channel numbers.

3.13.5.4.1 Receiver stability. The receiver when set to any channel, in the operating range, after 15 minute warmup and at any ambient temperature between -35 degrees F and +125 degrees F, shall not vary beyond plus or minus 0.002% ± 20 KC of that frequency during the next eight hours. The ambient may vary up to 54 degrees F within the above limits during the eight hour period.

3.13.6 IF bandwidth. The IF selectivity shall have a nominal 3 db bandwidth of 750 kc with the response down at least 60 db at ±2.1 MHZ or more from resonance.

3.13.7 Receiver protection. Receiver protection, in the form of electronic circuitry, shall be provided that when the transmitter has been tuned for optimum output it shall be possible to tune the receiver duplexer section through the transmitter frequency without damage to the receiver; the circuitry shall provide protection against external signals at any frequency with power lever up to +47 dBm under all conditions of receiver tuning. The protective circuitry shall not degrade the receiver noise figure greater than 0.750 dB.

3.13.8 Receiver frequency response. The overall frequency response of the receiver when a frequency modulated signal of constant deviation (100 KC peak) is applied to the receiver antenna terminal and measured at the video output terminal shall conform to Table III.



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TABLE III

| MOD. FREQUENCY Khz | REL. ATTENUATION dB, REFERENCE ATTENUATION AT 10 Khz | | |
|---|---|---|---|
| | MEDIAN | MINIMUM | MAXIMUM |
| .125 | 1.0 | .5 | 2.5 |
| .200 | .5 | 0 | 1.0 |
| .250 | .2 | -0.1 | .4 |
| 1.0 | 0 | -0.1 | .1 |
| 10 | REF | ----- | ----- |
| 30 | 0 | -0.1 | .1 |
| 60 | .15 | .05 | .25 |
| 120 | .6 | .2 | .9 |
| 180 | 1.3 | .8 | 1.8 |
| 240 | 2.3 | 1.8 | 2.8 |
| 360 | 5.1 | 4.4 | 5.8 |
| 480 | 9.0 | 8.2 | 10.0 |
| 600 | 14.0 | 12.5 | 15.5 |
| 720 | 20.2 | 18.4 | 22.6 |
| 840 | 27.5 | 25.3 | 31.0 |
| 960 | 36.0 | 33.0 | 40.0 |

3.13.9 Operational requirements. The radio equipment shall be capable of transmitting the PCM signals described in 3.13.10 through 3.13.16 and the FDM signals described in 3.13.17 through 3.13.25 meeting the performance requirements stated herein.

NOTE: All received rf signal levels are measured at the antenna terminals of the duplexer.

3.13.10 PCM operation. The radio set shall provide for the transmission of signals from multiplexers AN/TCC-44( ), AN/TCC-45( ), Pulse Code Modulation System, and Multiplexers TD-660( )/G and TD-754( )/G.



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These signals are briefly described below:

PCM Characteristics

| | | | |
|---|---|---|---|
| Number of channels | 6 | 12 | 24 |
| Signal Form | (Full-band, Unipolar Binary) | | (Bitenary combination of two 12 ch signals) |
| Bit rate | 288 | 576 | |
| Pulse width at Half amplitude | 3.472, usec | 1.736 usec | |

3.13.11 Input and output impedances (PCM).

a. The impedances at the video input to the transmitter shall be 91 ohms ±5% resistive to at least 480 KHZ and 91 ohms ± 15 % to at least 800 KHZ.

b. The output impedances of the receiver shall be 91 ohms ± 5% resistive to at least 1 MHZ for the regenerated 6/12 channel PCM output and 51 ohm ± 10% resistive for the 24 channel PCM output to at least 800 KHZ.

3.13.12 Frequency response (PCM). The overall frequency response between transmitter video input and receiver video output relative to the response at 10 kc shall conform to Table IV.



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TABLE IV

| FREQUENCY KHZ | SPECIFIED REL. ATTEN. dB REFERENCE 10 KHZ | | |
|---|---|---|---|
| | NOMINAL | MINIMUM | MAXIMUM |
| .125 | 2.0 | 1.0 | 3.0 |
| .200 | 1.0 | 0 | 2.0 |
| .250 | .4 | 0 | .8 |
| .500 | 0 | -.1 | +.1 |
| 1.0 | 0 | -.1 | +.1 |
| 10 | REF | --- | --- |
| 30 | 0 | -.1 | .2 |
| 60 | .2 | .1 | .4 |
| 120 | .7 | .5 | 1.0 |
| 180 | 1.6 | 1.3 | 2.2 |
| 240 | 2.9 | 2.5 | 4 |
| 360 | 6.5 | 5.7 | 8 |
| 480 | 11.7 | 10.2 | 13.5 |
| 600 | 18.2 | 16.0 | 21.0 |
| 720 | 26.2 | 23.1 | 30.5 |
| 840 | 35.7 | 31.7 | 41.0 |
| 960 | 46.7 | 41.2 | 52.5 |

3.13.13 PCM modulation requirements. The type of modulation utilized in the transmitter shall be frequency modulation. The rated peak deviation of the transmitter shall be ±300 kc. The modulation sensitivity of the transmitter shall be continuously variable with the maximum sensitivity being to deviate the transmitter plus and minus 300 kc peak with an input voltage of 1 volt peak to peak. The receiver regenerated output shall be 2 volts ± 10% when receiving a signal with peak deviation plus and minus 180 kc. The receiver unregenerated output shall be 3/4 volts ±1/4v peak to peak when receiving a signal with the rated peak deviation plus and minus 300 kc. Metering circuitry shall be provided in the equipment to enable the operator to perform the following in the absence of external test equipment:



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a. Adjust the peak deviation of the transmitter to plus and minus 180 kc and plus and minus 300 kc with an accuracy of plus 2.2 and minus 1.0 dB with inputs between 1 volt to 4 volts peak-to-peak.

b. Establish that the regenerated receiver output is 2 volts plus or minus 10% when receiving a signal with peak deviation of plus and minus 180 kc, and that the unregenerated receiver output is 3/4 volts plus or minus 1/4 volt peak to peak when receiving a signal with peak deviation of plus and minus 300 kc.

3.13.14 Signal to noise (PCM). The transmitter shall be tuned to any of its operating frequencies and shall be operated at rated rf power output. A 10 kc sinusoidal signal shall be applied at the video input terminals of the transmitter and the transmitter shall be adjusted for a peak deviation of plus and minus 180 KHZ. The receiver output, when the receiver input is coupled to the transmitter through sufficient attenuation, shall be 0.45v + 0.15v peak to peak at the unregenerated output terminals. Under these conditions the ratio of average detected signal power to the average noise power at the unregenerated video output terminals of the receiver as measured with a flat-weighted meter shall be at least 12 db when the received rf signal is -94 dbm, and at least 31 db when the received rf signal is -74 dbm. The transmitter shall be adjusted for rated peak deviation (plus and minus 300 KHZ) and the receiver unregenerated video output shall be 3/4 $\pm$ 1/4 volt peak to peak, the ratio shall be at least 16 db when the received rf signal is -94 dbm, and at least 35 db when the received signal is -74 dbm.

3.13.15 Distortion (PCM). The equipment shall be arranged as in 3.13.14 with the receiver coupled to the transmitter through sufficient attenuation that the rf input at the receiver input terminals (including duplexer) is minus 54 dbm. A sinusoidal signal at any frequency within the range 250 HZ to 240,000 HZ shall be applied at the video input terminals of the transmitter at a level of 2 volts peak to peak. The transmitter shall be adjusted for rated deviation and the receiver output shall be 3/4 plus or minus 1/4 volt peak to peak at the unregenerated video output terminals. Under these conditions, the total harmonic distortion measured at the unregenerated video output terminals shall not exceed a level 26 db below the fundamental.



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3.13.16 Pulse regeneration (PCM). The receiver output circuits shall contain means for generating new pulses where amplitude, timing and pulse shape are derived from the 6 channel or 12 channel signals specified in 3.13.10.

3.13.16.1 Regenerated PCM. The regneration shall eliminate all noise peaks less than 40% of the pulse amplitude for the range of unregenerated video output levels corresponding to input deviations of ± 145 KHZ peak minimum to ± 230 KHZ peak maximum. The regnerated signal amplitude shall be -0.1 volt ± 0.1 volt for the '1' condition and -2.0 volts ± 0.20 volts for the '0' condition, delivered into a load of 91 ohms ± 5%, with no reversal of phase from the transmitter input or unregnerated video output. Pulse width shall be full interval with 10% to 90% rise time 20 nsec minimum, 50 nsec maximum. The 50% point shall be 0 nsec minimum, 80 nsec maximum after the 50% point of the rising edge of the timing signal specified in 3.13.16.2.

3.13.16.2 Timing signal. Timing signals shall be extracted from the incoming pulse train and shall accompany the regenerated PCM to the next unit. The timing signal amplitude shall be 2.0 volts ± 0.2 dc isolated, delivered into a load of 91 ohms ± 5%. Pulse width at the 50% amplitude point shall be 90 nsec minimum, 150 nsec maximum. Rise time of the leading edge from the 10% point to the 90% point shall be 20 nsec minimum, 50 nsec maximum. Pulse rate shall be 576 KHZ for both the 6 channel and 12 channel signals.

3.13.16.3 Squelch and alarm. A squelch shall disable the regenerated PCM signal when the signal-to-noise ratio at the unregenerated video output is less than 7 db ± 2 db when measured as in 3.13.13, with a reference deviation of ± 180 KHZ peak. An audible and visible alarm shall operate in conjunction with the squelch to indicate locally, failure of the received signal from the remote station. A reversing switch shall be provided on the front panel such that regardless of the level of signal input to the receiver, the audible alarm can be disabled. The sensitivity of the squelch circuit shall be such that the point of operation which indicates a received signal (lamp extinguished) shall be -98 dbm ± 3 db.

3.13.17 FDM operation. The AN/GRC-103( ) shall provide for the transmission of signals from FDM Multiplexer AN/TCC-3( ). The FDM operational requirements specified herein shall be



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facilitiated by the basic radio set in conjunction with an FDM applique box, which shall be an accessory to the radio set. The FDM applique box shall contain hardware peculiar to the FDM required circuitry. The absence of the FDM applique shall in no way affect the PCM operation of the radio set.

3.13.18 Impedance (FDM). The nominal input impedance and the nominal output impedance at the FDM broadband terminal shall be 600 ohms over the band of 250 HZ to 20,000 HZ. The reflection coefficient at any modulation frequency between 500 HZ and 20,000 HZ shall not exceed 20%, and at any modulation frequency between 250 Hz and 500 Hz the reflection coefficient shall not exceed 40%. The degree of balance to ground at any modulation frequency between 250 HZ and 20,000 HZ shall be at least 30 db. These requirements shall be met for all settings of the input or output attenuator.

3.13.19 Input and output attenuator (FDM). An 11 position control consisting of a variable attenuation network shall be provided for the receive and transmit 600 ohm FDM terminals, and shall be accessible on the front panel of the FDM applique box. Attenuation shall decrease in 1 db steps with clockwise rotation of the control. The accuracy of attenuation shall be $\pm$.5 db for all settings of the control at all frequencies from 250 HZ to 20,000 HZ. The difference in attenuation between two successive steps shall be 1.0 db plus or minus 0.2 db.

3.13.20 Frequency response (FDM). The overall frequency response characteristics are to be met with any combination of transmitter, receiver and FDM applique box and shall be as follows:

a. Over the frequency range of 250 HZ to 4,000 HZ the response relative to 1 KHZ shall be within +1.0 db and -2.0 db.

b. Over the frequency range of 4,000 HZ to 20,000 HZ the response relative to 1 KHZ shall be within $\pm$1.0 db.

3.13.21 FDM modulation requirements. With a 1 KHZ sinusoidal signal of 191 mv applied to the transmitter video input, and the input control adjusted to give a reading of 50% on the panel meter, the transmitter peak deviation shall be between 65 KHZ and 97 KHZ. This requirement shall be met with the meter switch in the FDM position. When the FDM applique (Amplifier



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Audio Frequency AM-4324( )/GRC-103(V) is applied to the transmitter a ODBM, 1 KHZ sinusoidal signal across the TRANS MX terminals (600 ohms) of the applique shall produce a transmitter peak deviation between 65 KHZ and 97 KHZ. This condition shall be met with the input attenuator of the applique set to its midpoint, the meter switch of the transmitter set to FDM and the transmitter input control adjusted to give a reading of 50%.

3.13.21.1 FDM demodulation requirements. With a signal deviated 75 KHZ peak applied to the Receiver Antenna Terminal, the level measured across the video output connector terminated in 91 ohms shall be .189 V ± .02 V RMS. The meter of the receiver shall be in the green area with the switch in the FDM position. When the FDM applique is connected to the receiver the level measured across the 600 ohm REC MX terminals shall be 0 DBM ± 1 DB when a signal deviated 75 KHZ peak is applied to the Receiver Antenna Terminal. This condition shall be met with the output attenuator of the applique box set to its midpoint.

3.13.22 Pre-emphasis, de-emphasis network (FDM). A pre-emphasis network shall be incorporated in the transmit portion of the FDM applique box and a conjugate network shall be incorporated in the receive portion of the FDM. The networks shall be the same as in in the model.

3.13.23 Squelch and alarm (FDM). The FDM applique box shall include a squelch circuit which, when a radio frequency carrier is not being received, shall disable the FDM output. The sensitivity and control of the squelch circuit shall be such that it may be adjusted to operate for any signal power applied to the input terminals of the receiver in the range of minus 98 dbm to minus 74 dbm. A local audible and visible alarm shall operate in conjunction with the squelch to indicate failure of the received signal from the remote station. A switch, accessible on front panel, shall be provided which will disable the squelch. A reversing switch shall be provided on the front panel such that regardless of the level of signal input to the receiver or the setting of the squelch control the audible alarm can be disabled by means of the switch.



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3.13.24 Crosstalk (FDM). The crosstalk requirements of the FDM portion of this specification shall be fulfilled if the following test conditions are satisfied. For these measurements, the noise spectrum must be continuous and uniform to within plus or minus 1 db with reference to the level measured within a 4 KHZ portion at the center of the noise spectrum. The transmitter rf power output shall be coupled to the receiver through sufficient attenuation to give a power input of minus 50 dbm at the receiver input terminals. With modulation as specified in 3.13.20, the receive FDM output attenuator shall be adjusted for an overall 0 db gain between the transit FDM input terminals and receive FDM output terminals. At any carrier frequency in the range of the equipment, modulation shall be applied, consisting of a uniform random noise spectrum as follows:

| Frequency band | Mean power to 600 ohms FDM transmit terminal |
|---|---|
| a. 250 HZ to 16 KHZ | -2 dbm |
| b. 8 KHZ to 20 KHZ | -2 dbm |
| c. 0 to 4 KHZ | -5 dbm |

3.13.24.1 Crosstalk measurements. Under the above conditions the average power measured at the FDM output terminals falling in the following frequency bands shall be at least 42 db below the average power at the receiver audio terminals falling within any other 4 KHZ portion of the modulation spectrum.

a. 16 to 20 KHZ for test conditions 3.13.24a
b. 4 to 8 KHZ for test condition 3.13.24b
c. 4 to 8 KHZ for test condition 3.13.24c
d. 16 to 20 KHZ for test condition 3.13.24c

3.13.25 Signal-to-noise (FDM). At any assigned rf channel, the transmitter shall be coupled to the receiver. With modulation as specified in 3.13.21 FDM modulation requirements at rated peak deviations, the receive FDM output attenuator shall be adjusted for an overall 0 db gain between transmit FDM terminals and receive FDM output terminals. The ratio of average detected signal power to average noise power shall be measured at the FDM receive terminals, using a flat-weighted meter. The ratio in any 4 KHZ slot from 4 to 20 KHZ, with a sinusoidal modulating signal in that 4 KHZ slot, shall be at least:



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| Receiver rf input level | Signal-to-noise ratio |
|---|---|
| -94 dbm | 40 db |
| -74 dbm | 60 db |

3.13.26 Order wire operation. The following shall be provided by the radio equipment in conjunction with an order wire unit.

a. Provisions for order wire communications between a radio set and all other radio sets in the system by utilizing the portion of the transmission facilities below 4 KHZ, and providing order wire recovery.

b. Provisions for bridging a telephone handset to the transmission circuits of the radio set in both directions at a repeater station so that communications may be had in both directions simultaneously.

c. Provisions for signalling to or being signalled by the attendant at all other radio stations in the system.

d. Provisions shall be made to provide compatible order wire operation with Multiplexers TD-202( )/G, TD-204( )/G and TD-754( )/G.

3.13.26.1 Tone generator. A 1600 KHZ ± 4 HZ tone generator shall be incorporated into the O.W. unit to enable the requirements of 3.13.26c to be met. The tone generator shall be controlled by the operation of non-locking switch. A buzzer and lamp shall be provided in the unit which shall be energized when 1600 HZ tone is being received in the order wire channel. When the tone generator is activated by the non-locking switch the level measured at the TRANS "A" and TRANS "B" terminals of the order unit shall be -10 DBM ±1 DB.

3.13.26.2 Signal receiver detector sensitivity. When a -10 DBM ± 6 db 1600 HZ ± 4 HZ signal is applied to either the REC A or REC B terminals of the order wire unit, the signal lamp shall light and the buzzer sound.



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3.13.27 Receiver demodulation order wire. With an RF signal deviated ± 45 KHZ peak applied to the receiver antenna terminal, the level measured at the receiver order wire connector shall be -10 dbm ± 2 db.

3.13.28 Transmitter deviation (order wire). A 1000 HZ sinusoidal signal at a level of zero dbm applied to the microphone input terminals, of the order wire unit, shall produce a peak deviation of the transmitter between 39 and 58 KHZ. A 1000 HZ signal of - 10 dbm applied to the transmitter order wire connector shall produce a peak deviation of between 39 KHZ and 59 KHZ.

3.13.29 Signal-to-noise (order wire). The transmitter and receiver shall be connected together in a systems arrangement with sufficient attnenuation between them such that the power measured at the input terminals of the receiver is -94 dbm. With the transmitter modulated as in 3.13.28, the signal-to-noise ratio measured with FIA weighting at the local order wire terminals (receive phone terminals) shall be at least 39 db at any operating frequency in the presence of a normal 6/12 channel PCM signal and at least 40 db in the FDM mode of operation.

3.13.30 Peak limiting (order wire). The limiting action to be provided in the transmitter for the order wire shall be such as to compress test tone power inserted at the microphone terminals in Table V.

**Table V**

Limiting Characteristic, Insert Path

Measured between MIKE terminal and TRAN 'A' or TRAN 'B' terminals.

| INPUT POWER | COMPRESSION |
|---|---|
| -10 dbm | None |
| -5 dbm | None |
| 0 dbm | Less than 0.5 db |
| +5 dbm | 3.5 db ± 1 db |
| +10 dbm | 7.5 db ± 2 db |
| +15 dbm | 10.0 db ± 3 db |



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3.13.31 Frequency response (order wire system). The overall frequency response characteristic between the local microphone input and the phone output terminals of the order wire unit at a distant station shall be plus 1 to minus 3 db from 300 to 1800 HZ, referred to the 1 KHZ response at an rms level of zero dbm, when the equipment is employed in a full duplex PCM system.

3.13.32 Frequency response (order wire unit). The order wire unit RT-773( )/GRC-103 shall meet the following frequency (Table VI) response characteristics when measured with input levels specified in 3.13.36.

Table VI

OW Frequency Response

| PATH | FROM | TO | FREQUENCY RANGE | RESPONSE REF: 1 KHz |
|---|---|---|---|---|
| 'THRU' | REC 'A' | TRAN 'B' | 200 Hz -2.0 KHz | +0.5 db -1.0 db |
| 'THRU' | REC 'B' | TRAN 'A' | 200 Hz -2.0 KHz | +0.5 db -1.0 db |
| DROP | REC 'A' | PHONE terminal | 200 Hz -2.0 KHz | +0.5 db -1.5 db |
| DROP | REC 'B' | PHONE terminal | 200 Hz -2.0 KHz | +0.5 db -1.5 db |
| INSERT | MIKE terminal | TRAN 'A' | 200 Hz -2.0 KHz | +0.5 db -1.0 db |
| INSERT | MIKE terminal | TRAN 'B" | 200 Hz -2.0 KHz | +0.5 db -1.0 db |
| SIDETONE | MIKE terminal | PHONE terminal | 200 Hz -2.0 KHz | ± 1.0 db |



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3.13.33 Distortion (order wire system). With the equipment arranged as in 3.13.29 Signal-to-noise, the total harmonic distortion measured at the order wire output terminals shall not exceed 5%.

3.13.34 Distortion (order wire unit). The distortion of the order wire unit when measured with a 1000 HZ signal as specified in 3.13.32 shall be as follows:

| | | |
|---|---|---|
| 'THRU' PATHS | REC 'A' to TRAN 'B' | Less than 1% |
| | REC 'B' to TRAN 'A' | Less than 1% |
| INSERT PATHS | MIKE terminal to TRAN 'A' | Less than 2% |
| | MIKE terminal to TRAN 'B' | Less than 2% |
| DROP PATHS | REC 'A' to PHONE terminal | Less than 1% |
| | REC 'B' to PHONE terminal | Less than 1% |

3.13.35 Net loss (order wire system). The net loss between the local microphone input and the phone output at a distant station shall be 18 db ±3 db.

3.13.36 Order wire unit levels. The order wire unit levels when measured with a 1000 HZ signal shall conform to Table VII:

Table VII

| FROM | TO | OUTPUT LEVEL | INPUT LEVELS |
|---|---|---|---|
| REC 'A' | TRAN 'B' | -10 dbm ± 0.1 db (adjustable) | -10 dbm |
| REC 'B' | TRAN 'A' | -10 dbm ± 0.1 db (adjustable) | -10 dbm |
| REC 'A' | PHONE terminal | -18 dbm ± 1.0 db | -10 dbm |
| REC 'B' | PHONE terminal | -18 dbm ± 1.0 db | -10 dbm |
| MIKE terminal | TRAN 'A' | -10 dbm ± 1.0 db | 0 dbm |
| MIKE terminal | TRAN 'B' | -10 dbm ± 1.0 db | 0 dbm |
| MIKE terminal | PHONE terminal | -24 dbm ± 2.0 db (sidetone) | 0 dbm |
| REC 'A' | TRAN 'A' | -55 db ±10.0 db (crosstalk) | -10 dbm |
| REC 'B' | TRAN 'B' | -55 db ±10.0 db (crosstalk) | -10 dbm |



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3.13.37 Duplexers. The frequency ranges of the tunable duplexers shall correspond to the frequency bands of the radio set. The duplexers shall have minimum pass-band insertion loss. Insertion loss at the center frequency of the pass-band of the transmitter section shall not exceed 2 db over 90% of the operating band and shall not exceed 2.5 db over the remainder of the operating band.

a. Permit the operation of one antenna with a combination of one receiver and one transmitter on independent frequencies with a minimum separation of 10 MHz over the entire respective frequency [illegible] and bandwidths required to meet 3.13.2 (Operational requirements).

b. Permit the use of one antenna with either one transmitter or one receiver.

3.13.37.1 Voltage standing wave ratio. The voltage standing wave ratio measured at the antenna terminal of the duplexer, with the transmitter terminal terminated in a 50 ohm load and the receiver connected to the receiver connection, shall not exceed 2.0:1 at any transmitted frequency within the operating frequency range as specified in 3.13.37.

3.13.37.2 Tuning. The duplexers shall be continuously tunable over their respective frequency ranges. The tuning controls shall be the minimum consistent with meeting the requirements of this specification. To facilitate tuning, metering shall be incorporated to indicate power transmitted to the antenna from the duplexer and the power reflected from the antenna back to the duplexer. The controls shall be calibrated in channel numbers accurately enough to permit their tuning, without special equipment, to meet the operational requirements of this specification.

3.14 Antenna group.

3.14.1 Mast assembly. A complete mast assembly shall be provided capable of erecting and supporting the antenna and transmission line, described herein, to a height of fifty feet under the conditions of 3.14.4 and 3.14.5. The assembly shall consist of identical tubular mast sections (that form a continuous column when connected end to end) mast section launcher, guy assemblies, stakes, anchors, antenna adaptor, and tools.



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3.14.2 Mast sections. The basic mast section design concept shall be per the model.

3.14.3 Mast launcher. The mast launcher shall be per the model. The launcher base shall provide adequate resistance to the bearing and shear loads developed under conditions of 3.14.4 and 3.14.5, and a means to rotate and lock the mast sections in any azimuth orientation.

3.14.4 Erection. The mast shall be capable of erection to fifty (50) feet with the antenna and transmission line installed, in winds of twenty-five miles per hours with the launcher guyed and all other guy wires slack.

3.14.5 Operation. The mast assembly erected to fifty (50) feet with guy wires tensioned and antenna and transmission line installed, shall withstand the following conditions without permanent deformation, and maintain the antenna azimuth within $\pm 5^{\circ}$ of any initial setting.

a. Wind velocities up to 75 mph or;

b. Wind velocities up to 50 mph with a one-half inch radial ice load on the entire mast-antenna installation.

NOTE: The mast referenced in this specification is Mast AB-952( )/GRC-103(V) with Extension Kit, Mast MK-1009( )/GRC-103(V) to make a fifty (50) ft mast.

3.14.6 Symmetry. The ratio of the half-power (-3 db) E- and H- plane beamwidth shall not exceed 2.0:1 over the frequency range.

3.14.7 Squint. The gain of the antenna measured in the direction of the normal from the antenna shall not differ from the maximum gain measured at any frequency in the operating range by more than 0.5 db.

3.14.8 Minor-lobes. The ratio of major lobe of the antennas to any minor lobe shall be at least 12 db at any frequency in the operating range.



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3.14.9 Gain. The gain of the antenna relative to a dipole shall be at least 7 db at 220 MHz, 10 db at 400 MHz, 12.4 db at 1000 MHz, 14 db at 1350 MHz, 18 db at 1850 MHz, and 20 db at 2400 MHz.

3.14.10 Voltage standing wave ratio. The VSWR measured at the antenna terminals shall not exceed 1.75:1 over the operating frequency range.

3.14.11 Impedance. The nominal input impedance of each antenna shall be 50.0 ohms.

3.15 Systems level stability. The transmitter and receiver shall be connected through an attenuator with sufficient attenuation such that the power input to the receiver input terminals is minus 60 dbm. With the transmitter modulated as described in 3.13.13 (PCM modulation requirements) and the overall system adjusted for nominal gain, the following conditions shall be met: The system gain shall not vary more than +1 db in any eight hour period between the 91 ohm video input and 51 ohm video output terminals of the receiver when the received signal is varied between -95 dbm and -45 dbm under any combination of temperature, humidity or line voltage variation as specified herein.

3.15.1 Mutual interference. A transmitter shall be tuned to any channel in its band and coupled to an antenna through the duplexer. With a receiver connected to the same antenna through the duplexer, there shall be a maximum of eight channels unusable by the receiver, exclusive of the channels in a ±16 MHz band centered around the transmitter frequency. An unusable channel shall be defined as a channel on which the receiver cannot receive a remote signal of level -94 dbm and meet the requirements of 3.13.9.

3.16 Power requirements. The equipment shall operate from a 115 volt rms, 47.5 hertz to 420 hertz, single phase source, having a voltage variation of plus or minus 10%. The total power drain for a transmitter and receiver shall not exceed 400 watts at normal operating conditions.

3.17 Packaging. The equipment shall be packaged in cases in accordance with the model. Nothing shall be done electrically or mechanically to preclude the use of the other plug in assemblies for operation in other bands.



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3.18 Physical characteristics.

3.18.1 Size. The equipment cases housing the individual radio set components shall not exceed the following dimensions:

Transmitter (with integral Power Supply and any one tuning unit) depth - 12 inches, width-17-3/8 inches, height-8.5 inches.

Receiver (with integral Power Supply and any one tuning unit) depth-12 inches, width-17-3/8 inches, height-8.5 inches.

3.18.2 Weight. The weight of the radio set when installed in its cases shall not exceed the following:

Transmitter (with integral Power Supply and any one tuning unit) -65 pounds.

Receiver (with integral Power Supply and any one tuning unit) 65 pounds

Receiver - Transmitter, Order Wire RT-773( )/GRC-103(V) - 7.6 pounds.

3.19 Metering. The metering provided shall be the same as in the model.

3.20 Safety measures.

3.20.1 Mechanical. The equipment shall be such that, when set up for operation, accidental contact by personnel with moving parts such as gears, fans and belts will be prevented. Projecting and overhanging edges shall be held to a minimum. Edges and corners of enclosures shall be rounded.

3.20.2 Electrical.

3.20.2.1 Low voltage. With the equipment assembled and set up for operation, personnel shall be protected from contact with potentials in excess of 30 volts rms ac or dc to ground, chassis, or frame, including potentials on charged capacitors.. However, the potential on an antenna, counterpoise, or transmission line, including their terminals, may exceed 30 volts unless the transmitted energy is used for electrical power.



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3.20.2.2 High voltage. Safety measures shall be taken against high voltage, as follows:

3.20.2.2.1 Interlocks: Interlocks shall be provided where necessar to prevent access to the interior of the equipment, or compartments thereof, without removal of all voltages exceeding 150 volts between any point and ground. Interlocks shall conform to MIL-P-11268.

3.20.2.3 Warning label. Where a voltage of 200 or more volts nominal value is exposed during maintenance a warning label shall be provided. The label shall read "DANGER XXX VOLTS WHEN COVER REMOVED," in aluminum or white lettering on a red background, where XXX shall read the nominal voltage.

3.20.2.4 Terminal covers. Terminals strips having potentials in excess of 30 volts rms ac or dc which might be accidentally contacted by the operator during cover off operation shall have a plastic cover which shall be slotted to allow instrument probes to be inserted to the contacts or terminals. The barrier cover shall be marked with a warning label "EXPOSED TERMINALS DANGER XXX VOLTS," where XXX shall read the nominal voltage.

3.20.2.5 Circuit protection. Equipment shall be protected by circuit breakers connected directly to all input terminals. Protective devices shall be conveniently arranged and located for safe and easy removal or resetting.

3.20.2.6 Grounding. The primary power cable shall carry a third ground wire connecting the metal chassis portions of all interconnected equipment, in addition to the two energized wires (one of which may be grounded during operation).

3.21. Service conditions.

3.21.1 Temperature.

3.21.1.1 Low. During testing as specified in 4.7.1.1 the equipment shall meet the performance specified in 4.9.1. After testing, the equipment shall meet full specification performance for these measurments as specified in 4.9.1.



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3.21.1.2 High. During testing as specified in 4.7.1.2 the equipment shall meet the performance specified in 4.9.1. After testing, the equipment shall meet full specification performance for these measurements as specified in 4.9.1.

3.21.2 Moisture resistance (humidity). During testing as specified in 4.7.2 the equipment shall meet the performance requirements of 4.9.1. After testing, the equipment shall meet full specification performance for these measurements as specified in 4.9.1.

3.21.3 Elevation (Altitude). The equipment shall meet full specification performance of 4.9.2 during and after testing as specified in 4.7.3.

3.21.4 Vibration, bounce and shock. The equipment shall be capable of meeting the requirements of Table VIII.



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TABLE VIII

Bounce and Shock

| Paragraph heading | Test Para. | Performance after test |
|---|---|---|
| Bounce | 4.7.4.1 | Specified performance. (Notes A and B) No physical damage. |
| Shock; ballistic | 4.7.4.2 | Operable (Note B). Any physical damage shall be minor only. |
| Shock; bench-handling | 4.7.4.3 | Specified performance (Note B). No physical damage. |

NOTE A: With the equipment shock-mounts (if any) blocked, the amplitude of vibration of any part, subassembly, or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 hertz.

NOTE B: The equipment shall meet specified performance for the following:

a. Paragraph 4.2 and 4.5 of MIL-R-55656/4, except test channels shall be 40, 179, 180 and 400 at 115 volts, 60 Hz.

b. Paragraph 4.5 of MIL-R-55656/3 except test channels shall be 43, 124 and 370.

c. Paragraph 4.2.2 of MIL-R-55656/5.

d. Paragraph 4.1 of MIL-R-55656/5 except test channels shall be 40, 179, 180 and 400.

e. Paragraph 4.5.2 of MIL-R-55656/5.

f. Paragraph 4.10 of MIL-R-55656(EL).



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3.21.5 Dripproof. The equipment shall meet the requirements of 4.9.1 when tested in accordance with 4.7.5.

3.21.6 Salt fog. After testing as specified in 4.7.6 the equipment shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

3.21.7 Sand and dust. The equipment shall meet the requirements of 4.10 when tested in accordance with 4.7.7.

3.21.8 Fungus. The equipment (including accessories) shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.7.8. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

3.22 Reliability. The reliability of the equipment shall provide not less than a mean-time-between-failure (MTBF) of 1200 hours assuming an exponential distribution. The reliability shall be demonstrated in accordance with the test procedure of 4.12.2.

3.23 Electromagnetic interference characteristics. The equipment shall meet the electromagnetic interference requirements of MIL-STD-461A, Table A-1 as defined herein. If the contractor considers it desirable to modify the equipment proper in order to facilitate conformance with the specified emanation and susceptibility limits, he shall obtain approval from the contracting officer before proceeding with modifications. The following Class CE requirements apply to the GRC-103.

| | | | | |
|---|---|---|---|---|
| CE06 key-up only | CS01<br>CS06 | RE02 | RS03 | A-CE1<br>A-CS1 |

3.24 Workmanship. The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:



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In MIL-P-11268:

| | |
|---|---|
| General requirements for plastic material and plastic parts. | Cleaning of units. |
| Wiring and cabling, including: | General requirements for securing of parts. |
| Slack. | Process for welding. |
| Protection. | Cleaning after welding. |
| Clearance. | Controls. |
| Splicing. | Self-tapping screws. |
| Connections, general. | |
| Grounding, general. | |
| Shielding on wire and cable. | Cleaning prior to welding. |
| Soldering and brazing. | |

3.25 Preconditioning.

3.25.1 Bounce Preconditioning. The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6.1 (Also see 4.5).

3.25.2 Electrical Preconditioning. (See 4.6.2) Prior to the contractor's electrical assembly testing, all units of Radio Set AN/GRC-103( )(V) (including all spares) shall be electrically preconditioned for a minimum of 48 hours. The preconditioning shall consist of the following:

a. Normal or simulated operating signals shall be applied to the equipment during the operational preconditioning and periodic checks performed to detect any failures that may occur. During this operational preconditioning, the units shall be periodically de-energized and re-energized individually for a minimum number of six (6) times. (Off-time: 2 minutes; On-time interval: 2 hours)

b. The equipment shall have no less than 48 hours of operational preconditioning nor more than 144 hours of operational preconditioning (exclusive of production test time) when used as a vehicle for preconditioning power supplies plug-in subassemblies (including all spares).

3.26 Technical literature, tools, and running spare parts. Technical literature, tools, and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows.

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3).

(b) Inspection covered by subsidiary documents. (See 4.4).

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5).

(2) Quality conformance inspection of preparation for delivery. (See 4.22).



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4.3 Preproduction inspection. This inspection will be performed by the contractor unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in Table IX, the inspection specified in the subsidiary documents covering the items listed in 4.4 and the inspection specified for group A, group B and group C (see Tables X, XI, and XII, respectively).

Table IX

Preproduction Inspection

| Inspection (For additional preproduction inspection see 4.3) | Req para | Insp para |
|---|---|---|
| Temperature | 3.21.1 | 4.7.1 |
| Relative Humidity | 3.21.2 | 4.7.2 |
| Elevation | 3.21.3 | 4.7.3 |
| Dripproof | 3.21.5 | 4.7.5 |
| Electromagnetic Compatibility | 3.23 | 4.21 |
| Vibration, internal | 3.21.4 | 4.7.4.1 |
| Bounce | 3.21.4 | 4.7.4.1 |
| Shock, Ballistic | 3.21.4 | 4.7.4.2 |
| Shock, Bench-handling | 3.21.4 | 4.7.4.3 |
| Salt Fog | 3.21.6 | 4.7.6 |
| Sand and Dust | 3.21.7 | 4.7.7 |
| Fungus | 3.21.8 | 4.7.8 |
| Reliability | 3.22 | 4.12 |
| Power Line, Voltage and Frequency | 3.16 | 4.11 |
| Antenna and Mast | 3.14 | 4.13 |
| Electromagnetic Interference | 3.13.4.5<br>3.13.5.3<br>3.23 | 4.21 |



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4.4 Quality Conformance inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the Quality Conformance before preparation for delivery:

| ITEM | WHERE REQUIRED |
|---|---|
| Finish | 3.5 |
| Marking | 3.6 |
| Workmanship | 3.24 |
| Parts Materials & Processes | 3.4 |

4.5 Quality conformance inspection of equipment before preparation for delivery. Each unit of each lot of equipment shall be subjected to bounce preconditioning (See 3.25) prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of bounce preconditioning shall be classified as a defect and recorded as such. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government -- at its discretion -- may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirement. Test equipment for Government verification inspection shall be made available by the contractors. Each unit of each lot shall be subjected to final Operational Check 4.5.1.1 immediately prior to packaging.

4.5.1 Group A Inspection. Each unit of each lot of equipment shall be inspected for conformance to all inspections and test requirements of Table X. Each lot will be subject to verification, utilizing the procedures of Military Standard MIL-STD-105, and the AQLs indicated in Table X. Group A inspection shall be performed in any order which is satisfactory to the Government.



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TABLE X - Group A Inspection

| Inspection | Req para | Insp para | AQL* Major | AQL* Minor |
|---|---|---|---|---|
| (a) Visual and Mechanical | 3.4, 3.5 | 4.19 | | |
| Receiver, Radio (1RE1) R-1329 (P)/GRC-103(V) | 3.6, 3.24 | | | |
| Synthesizer (1RE1A2) | | | 1% | 4% |
| Panel, Monitor (1RE1A1) | | | 1% | 4% |
| Power Supply (1RE1PS1) | | | 1% | 4% |
| Restorer, Pulse Form (1RE1A3) | | | 1% | 4% |
| Limiter-Discriminator (1RE1A4) | | | 1% | 4% |
| Amplifier-Frequency Multiplier (1RE1A5) | | | 1% | 4% |
| Amplifier, Video (1RE1AR1) | | | 1% | 4% |
| Amplifier, Intermediate Freq. (1RE1AR2) | | | 1% | 4% |
| Case, Receiver (1A2) (CY-4638/GRC-103(V) | | | 1% | 4% |
| Distribution Box (1A2A1) | | | 1% | 4% |
| Amplifier - Converter(2A1) AM-4316( )/GRC-103(V) | 3.4, 3.5 3.6, 3.24 | 4.19 | | |
| Duplexer (2A1A1) | | | 1% | 4% |
| Converter, Frequency, Electronic (2A1A2) | | | 1% | 4% |
| Amplifier, Radio Frequency (2A1AR1) | | | 1% | 4% |
| Control-Indicator (2A2) | | | 1% | 4% |
| Electronic Switch (2A4) | | | 1% | 4% |
| Power Supply (2PS1) | | | 1% | 4% |
| Transmitter, Radio (5TR1) T-983 (P) ( )/GRC-103(V) | 3.4, 3.5 3.6, 3.24 | 4.19 | | |
| Synthesizer (5TR1A2) | | | 1% | 4% |
| Panel, Monitor (5TR1A1) | | | 1% | 4% |
| Power Supply (5TR1PS1) | | | 1% | 4% |
| Control, Alarm (5TR1A3) | | | 1% | 4% |
| Amplifier-Frequency Multiplier (5TR1A4) | | | 1% | 4% |
| Amplifier-Monitor (5TR1A5) | | | 1% | 4% |
| Case, Transmitter (5A2) (CY-4637( )/GRC-103(V) | | | 1% | 4% |
| Distribution Box (5A2A1) | | | 1% | 4% |



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4.5.1 TABLE X (Cont'd)

| Inspection | Req Para | Insp Para | AQL* Major | AQL* Minor |
|---|---|---|---|---|
| Amplifier-Frequency Multiplier (6) (AM-4320( )/GRC-103(V) | 3.4, 3.5 3.6, 3.24 | 4.19 | | |
| Amplifier, Radio Frequency (6AR1) | | | 1% | 4% |
| Electronic Switch (6A1) | | | 1% | 4% |
| Freqeuncy Multiplier Assy. (6A2) | | | 1% | 4% |
| Control-Indicator (6A3) | | | 1% | 4% |
| Control, RF Power Level (6A5) | | | 1% | 4% |
| Receiver-Transmitter, Order Wire (9) (RT-773( )/GRC-103(V) | 3.4, 3.5 3.6, 3.24 | 4.19 | | |
| Panel, Distribution (9A1) | | | 1% | 4% |
| Panel, Monitor (9A2) | | | 1% | 4% |
| Converter, Telephone Signal (9A3) | | | 1% | 4% |
| Amplifier Assembly (9A4) | | | 1% | 4% |
| Case, Recvr.-Trans., Order Wire (9MP1) | | | 1% | 4% |
| Dummy Load, Electrical (16) (DA-437( )/GRC-103(V) | 3.4, 3.5 3.6, 3.24 | 4.19 | 1% | 4% |
| Antenna (11) (AS-1852( )/GRC-103(V) | 3.24 | 4.19 | | |
| Reflector, Antenna (11A1) | | | 1% | 4% |
| Antenna Element (11A2) | | | 1% | 4% |
| Mast (12) (Mast AB-952( )/GRC-103(V) | 3.24 | 4.19 | 1% | 4% |
| Launcher | 3.24 | 4.19 | 1% | 4% |
| Mast Section | 3.24 | 4.19 | 1% | 4% |
| Accessory Bag | 3.24 | 4.19 | 1% | 4% |
| Guy Assy | 3.24 | 4.19 | 1% | 4% |
| Guy Ring | 3.24 | 4.19 | 1% | 4% |
| Base Pins | 3.24 | 4.19 | 1% | 4% |
| Ant Adaptor | 3.24 | 4.19 | 1% | 4% |
| Stakes | 3.24 | 4.19 | 1% | 4% |
| Hammer | 3.24 | 4.19 | 1% | 4% |
| Extension Kit | 3.1.2.9 3.24 | 4.19 | 1% | 4% |



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4.5.2 TABLE X (Cont'd)

| Inspection | Req para | Insp para | AQL* Major | AQL* Minor |
|---|---|---|---|---|
| (b) Electrical (Subassemblies) | | | | |
| Receiver, Radio (1RE1) | | | | |
| Synthesizer (1RE1A2) | 3.12 | 4.18.5 | | |
| Frequency Check<br>Synch. Alarm<br>Deviation<br>Power Output | | | 1% for the group | |
| Panel, Monitor (1RE1A1) | | | | |
| Continuity Check | 3.12 | 4.18.61 | 1% | |
| Power Supply (1RE1PS1) | 3.12 | 4.18.11 | | |
| Initial Calibration<br>Meter Check<br>Regulation<br>Load & Regulation<br>Unregulated Supply<br>Ripple Measurement (Unreg.)<br>Ripple Measurement (Reg.)<br>High Frq. Instability | | | 1% for the group | |
| Restorer, Pulse Form (1RE1A3) | 3.12 | 4.18.6 | | |
| Timing Pulse<br>PCM Regenerator<br>OW Noise Test<br>OW Distortion | | | 1% for the group | |



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4.5.1 TABLE X (Cont'd)

| Inspection | Req para | Insp para | AQL* Major | Minor |
|---|---|---|---|---|
| Limiter-Discriminator (1RE1A4) | 3.12 | 4.18.7 | | |
| Demod. Linearity<br>Deviation Sensitivity<br>Frequency Response | | | 1% for the group | |
| Amplifier-Frequency Multiplier (1RE1A5) | 3.12 | 4.18.8 | | |
| Harmonic Level | | | 1% | |
| Amplifier, Video (1RE1AR1) | 3.12 | 4.18.9 | | |
| Gain<br>24CH PCM Adjust<br>Low Signal Alarm<br>Frequency Response<br>Distortion<br>Continuity Check | | | 1% for the group | |
| Amplifier, Intermediate Frequency (1RE1AR2) | 3.12 | 4.18.10 | | |
| Gain (without AGC)<br>Frequency Response<br>AGC Limited Output Level | | | 1% for the group | |
| Case, Receiver (1A2) (CY-4638( )/GRC-103(V) | | | 1% | |
| Distribution Box (1A2A1) | 3.12 | 4.18.61 | | |
| Continuity check | | | | |
| Amplifier-Converter (2) (AM-4316( )/GRC-103(V) | | | | |
| Tuner, Radio Frequency (2A1) | 3.12 | 4.18.19 | 1% | |



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4.5.1 TABLE X (Cont'd

| Inspection | Req para | Insp para | AQL* Major | Minor |
|---|---|---|---|---|
| Duplexer(2A1A1) | 3.12 | 4.18.24 | 1% for the group | |
| Adjustment of Lead Lengths | | | | |
| Receiver Channel Calibration | | | | |
| Transmitter Channel Calibration | | | | |
| VSWR Bandwidth | | | 1% for the group | |
| Receiver Path Insertion Loss | | | | |
| Receiver Path Ripple | | | | |
| Transmitter Path Insertion Loss | | | | |
| Transmitter Path Ripple | | | | |
| Local Oscillator Signal Tracking | | | | |
| Converter, Frequency Electronic (2A1A2) | 3.12 | 4.18.25 | | |
| Frequency Tracking | | | 1% for the group | |
| VSWR | | | | |
| Local Oscillator Calibration | | | | |
| Noise Figure | | | | |
| Amplifier, Radio Frequency (2A1AR1) | 3.12 | 4.18.27 | | |
| Gain and Bandwidth | | | 1% for the group | |
| VSWR | | | | |
| Noise Figure | | | | |
| Control-Indicator (2A2) | 3.12 | 4.18.20 | | |
| Coding Check | | | | |
| Electronic Switch (2A4) | 3.12 | 4.18.22 | | |
| VSWR | | | 1% for the group | |
| Insertion Loss | | | | |
| Isolation Measurement | | | | |
| System | | | | |



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4.5.1 TABLE X (Cont'd)

| Inspection | Req para | Insp para | AQL* Major | AQL* Minor |
|---|---|---|---|---|
| Power Supply (2PS1) | 3.12 | 4.18.23 | | |
| Meter Check<br>Overload Alarm<br>Ripple | | | 1% for the group | |
| Transmitter, Radio (5TR1)<br>Panel, Monitor (5TR1A1)<br>Continuity Check | 3.12 | 4.18.61 | | |
| Synthesizer (5TR1A2) | 3.12 | 4.18.5 | | |
| Frequency Check<br>Synch. Alarm<br>Deviation<br>Power Output | | | 1% for the group | |
| Control, Alarm (5TR1A3) | 3.12 | 4.18.40 | | |
| Low Power Alarm and Backlash Adjustment | | | | |
| Amplifier-Frequency Multiplier (5TR1A4) | 3.12 | 4.18.41 | | |
| Frequency Response<br>Output Level<br>Power Output (with ALC)<br>Harmonic Suppression<br>Input VSWR | | | 1% for the group | |
| Amplifier-Monitor (5TR1A5) | 3.12 | 4.10.42 | | |
| Operational Check<br>Metering Check<br>Frequency Response | | | 1% for the group | |



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4.5.1 TABLE X (Cont'd)

| Inspection | Req para | Insp para | AQL* Major | AQL* Minor |
|---|---|---|---|---|
| Power Supply (5TR1PS1) | 3.12 | 4.18.43 | | |
| Insulation and Resistance<br>Preliminary Check<br>Meter Reading<br>Input Rectification<br>High Tension<br>Rise Time and Frequency Measurement at 5KHZ Converter<br>Rise Time and Frequency Measurement at 400 Hz Converter<br>26V dc Unreg. Supply<br>Filament Supply<br>28V dc/12V dc Reg. Supply<br>High Frequency Instability<br>Constant Current<br>High Tension Metering Circuit Calibration<br>5kHz Transistor Saturation<br>Blower Voltage and Alarm Supply<br>Hum Measurement (120 Hz)<br>Spike Measurement (5kHz)<br>Input AC Current<br>Short Circuit Protection Check<br>Starting Circuit Check | | | 1% for the group | |
| Case, Transmitter (5A2) (CY-4637( )/GRC-103(V) | | | | |
| Distribution Box (5A2A1) Continuity Check | 3.12 | 4.18.61 | 1% | |
| Amplifier-Frequency Multiplier (6) (AM-4320( )/GRC-103(V) | | | | |
| Amplifier, Radio Frequency (6AR1) | 3.12 | 4.18.51 | | |



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4.5.1 TABLE X (Cont'd)

| Inspection | Reg Para | Insp Para | AQL * Major | AQL * Minor |
|---|---|---|---|---|
| Drive Power Calibration<br>Output Amplifier-Power Output<br>Output Amplifier-Input Matching<br>Driver Amplifier-Power Output<br>Driver Amplifier-Input Matching<br>Complete Amplifier-Tuning and Power Output | | | 1% for the group | |
| Electronic Switch (6A1)<br>Operational Check<br>VSWR<br>Insertion Loss<br>Rejection Ratio<br>Power Monitor | 3.12 | 4.18.47 | 1% for the group | |
| Frequency Multiplier Assembly (6A2) | 3.12 | 4.18.48 | | |
| Tuning Procedure Band 1B<br>Tuning Procedure Band 1A | | | 1% | |
| Control-Indicator (6A3)<br>Coding Check | 3.12 | 4.18.49 | 1% | |
| Control, RF Power Level (6A5) | 3.12 | 4.18.50 | | |
| Input VSWR (Zero Control Voltage)<br>Input VSWR (Operating Voltage)<br>Input VSWR (Diodes Conducting)<br>Insertion Loss | | | 1% for the group | |
| Receiver-Transmitter Order Wire (9) (RT-773( )/GRC-103(V)) | | | | |
| Panel, Distribution (9A1) | 3.12 | 4.18.56 | | |
| Point-to-point Resistance | | | 1% | |



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4.5.1 TABLE X (Cont'd)

| Inspection | Reg Para | Insp Para | AIQ * Major | Minor |
|---|---|---|---|---|
| Panel, Monitor (9A2) | 3.12 | 4.18.57 | | |
| Point-to-point Resistance | | | 1% | |
| Converter, Telephone Signal (9A3) | 3.12 | 4.18.58 | | |
| Oscillator Frequency<br>Signal Receiver Selectivity<br>Signal Receiver Sensitivity | | | 1%<br>for the<br>group | |
| Amplifier Assembly (9A4) | 3.12 | 4.18.59 | | |
| Thru Paths<br>Power Gain<br>Frequency Response<br>Harmonic Distortion<br>Crosstalk | | | 1%<br>for the<br>group | |
| Insert Paths<br>Power Gain<br>Frequency Response<br>Harmonic Distortion<br>Signal Compression | | | 1%<br>for the<br>group | |
| Drop Paths<br>Power Gain<br>Frequency Response<br>Harmonic Distortion<br>Sidetone Level | | | 1%<br>for the<br>group | |
| (c) Electrical (Main Assemblies) | | | | |
| Receiver, Radio (R-1329(P)( )/GRC-103(V) with Amplifier-Converter (AM-4316( )/GRC-103(V)) | | | | |



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4.5.1 TABLE X (contd)

| Inspection | Req Para | Insp Para | AQL * Major | Minor |
|---|---|---|---|---|
| Frequency Range | 3.13.2 | 4.14 | | |
| Noise Figure | 3.13.5.1 | 4.14 | 1% | |
| Setability | 3.13.5.4 | 4.14 | for the | |
| IF Bandwidth | 3.13.6 | 4.14 | group | |
| Receiver Protection | 3.13.7 | 4.14 | | |
| Frequency Response | 3.13.8 | 4.14 | | |
| VSWR (Duplexers) | 3.13.37.1 | 4.14 | | |
| Transmitter, Radio (T-983(P)( )/ GRC-103(V) with Amplifier-Frequency Multiplier (AM-4329( )/ GRC-103(V) | | | | |
| Frequency Range | 3.13.2 | 4.15 | | |
| Power Output | 3.13.4.1 | 4.15 | 1% | |
| Power Dissipation | 3.13.4.2 | 4.15 | for the | |
| Setability | 3.13.4.4 | 4.15 | group | |
| Low Power Alarm | 3.13.4.6 | 4.15 | | |
| Receiver-Transmitter, Order Wire (RT-773( )/GRC-103(V)) | | | | |
| Tone Generator | 3.13.26.1 | 4.16 | 1% | |
| Deviation | 3.13.28 | 4.16 | for the | |
| Sidetone | 3.13.36 | 4.16 | group | |
| Peak Limiting | 3.13.30 | 4.16 | | |
| Dummy Load, Electrical (DA-437( )/GRC-103(V)) | | | | |
| VSWR | 3.13.48 | 4.18.60 | 1% | |
| Impedance | 3.13.48 | 4.18.60 | for the | |
| Dissipation | 3.13.48 | 4.18.60 | group | |
| Antenna (AS-1852( )/GRC-103(V)) | | | | |
| VSWR | 3.14.10 | 4.13 | 1% | |
| Impedance | 3.14.11 | 4.13 | for the group | |



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4.5.1.1 Final operational check - Radio Set AN/GRC-103( )(V)). The final inspection of the radio set immediately prior to packaging for shipment shall be an operational check of the units set up in a system The operational check shall consist of voice communications and order wire ringing through the system plus a visual check of the metering facilities for normal readings. The equipment shall be operated, as a minimum, at transmitter channels 40,179, and 180 and receiver channels 43,124, and 370.

4.5.2 Group B inspection. This inspection, including sampling, shall conform to Table XI and the special procedures for small sample inspection of Standard MIL-STD -105. The inspection level shall be S-3. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met Group A inspection.

TABLE XI - Group B Inspection

| Inspection | Req Para | Insp Para | AQL * (Major) |
|---|---|---|---|
| (a) Electrical (Main Assemblies Receiver, Radio (R-1329(P)( )/GRC-103(V)with Amplifier-Converter (AM-4316 ( )/GRC-103(V)) | | | |
| Output Impedance | 3.13.11b | 4.14 | 6.5% |
| Squelch and Alarm | 3.13.16.3 | 4.14 | for the |
| Power Requirement | 3.16 | 4.14 | group |
| Transmitter, Radio (T-983(P) ( )/GRC-103(V) with Amplifier-Frequency Multiplier (AM-4320( ) /GRC-103(V)) | | | |
| Input Impedance | 3.13.11a | 4.15 | 6.5% |
| Power Requirement | 3.16 | 4.15 | for the group |



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4.5.2.1 Interchangeablilty. Electrical and mechanical interchangeability shall be demonstrated by disassembling all normally removable modules from two units that have been subjected to and met all other Group B tests, interchanging approximately one-half the modules and reassembling. The main frame assemblies (fixed and tuning heads) shall be interchanged between cases. The units shall then be operated and measurements made of transmitter power output and receiver noise quieting at two frequencies in the band. The units shall meet the requirements of paragraphs 3.13.4.1 and 3.13.5.1 of this specification.

TABLE XI - Group B Inspection (contd)

| Inspection | Req Para | Insp Para | AQL* (Major) |
|---|---|---|---|
| (b) Electrical (System) Radio Set AN/GRC-103( )/(V)1 | | | |
| PCM Operation | 3.13.10 | 4.17 | 6.5% for the group |
| Frequency Response | 3.13.12 | 4.17 | |
| PCM Modulation | 3.13.13 | 4.17 | |
| PCM Signal-to-Noise | 3.13.14 | 4.17 | |
| PCM Distortion | 3.13.15 | 4.17 | |
| Pulse Regneraration | 3.13.16.1 | 4.17 | |
| Timing Signal | 3.13.16.2 | 4.17 | |
| Order Wire Signal-to-Noise | 3.13.29 | 4.17 | |
| Order Wire Distortion | 3.13.34 | 4.17 | |
| Duplexer | 3.13.37 | 4.17 | |
| Duplexer Tuning | 3.13.37.2 | 4.17 | |
| Order Wire Frequency Response | 3.13.31 | 4.17 | |
| Order Wire Net Loss | 3.13.35 | 4.17 | |



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4.5.3 Group C Inspection. This inspection shall be as listed in Table XII and shall be performed on units that have been subjected to and met Group A and Group B inspections. Units selected for each Group C inspection shall be selected without regard to their quality. The inspections listed in Table XII shall be performed on one unit for each 100 units produced. The First Article tests shall be considered as satisfying the requirement for the first 100 units produced.

TABLE XII

Group C Inspection

| Inspection | Req Para | Insp Para |
|---|---|---|
| (a) Mechanical Mast AB-952( )/GRC-103(V) (See Note 1) | | |
| Erection | 3.14.4 | 4.13 |
| Operation | 3.14.5 | 4.13 |
| (b) Electrical (System) Radio Set AN/GRC-103(V) (Perform for each 100 units produced) (See Note 2) | | 4.9.1 |
| Temperature and Humidity | 3.21.2 | 4.9.1 |
| Transmitter Frequency Stability | 3.13.4.3 | 4.9.1 |
| Transmitter Power Output | 3.13.4.1 | 4.9.1 |
| Transmitter Modulation | 3.13.13 | 4.9.1 |
| Receiver Noise Figure | 3.13.5.1 | 4.9.1 |
| Duplexer Insertion Loss | 3.13.37 | 4.9.1 |
| PCM Frequency Response | 3.13.12 | 4.9.1 |
| PCM Squelch and Alarm | 3.13.16.3 | 4.9.1 |
| PCM Net Gain and Stability | 3.15 | 4.9.1 |
| Order Wire Signal to Noise | 3.13.29 | 4.9.1 |
| Shock Bench Handling | 3.21.4 | 4.9.1 |

NOTES:

1. Perform erection and vibration test once for each 100 units produced and temperature and relative humidity tests bi-monthly.

2. Perform electromagnetic interference and bounce (vibration) tests once for each 500 units produced.



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4.5.3.1 Noncompliance. The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and, (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C Inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Group A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.4 Reinspection of Conforming Group B and Group C Sample Units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C Inspection may be accepted on contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Preconditioning.

4.6.1 Bounce preconditioning. The unit, with shock mounts (if any) removed or blocked, shall be placed in its normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to acceptance inspection. All equipment delivered on the contract or order shall be subjected to the requirements of this paragraph. If an equipment fails any test because of the bounce preconditioning, the defect applies to that test.



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4.6.2 Electrical pre-conditioning inspection. Electrical pre-conditioning inspection shall meet the requirements of 3.25.2 (Electrical preconditioning).

a. The unit, assembly, sub-assembly or module shall be considered to have been successfully electrically preconditioned when they have individually passed a cumulative 48-hour-operating preconditioning period without failure. If during the 48-hour-operating preconditioning period a failure occurs, repair may be made using preconditioned sub-assemblies or modules and the pre-conditioning continued without the restart of a new 48-hour preconditioning period. A sub-assembly or module failing operating preconditioning shall, after repair, be resubmitted to and pass the entire 48-hour-operating preconditioning period prior to final electrical assembly tests by the contractor.

b. The contractor shall perform tests approved by the government to demonstrate normal operation of the equipment during the periodic operational tests. A test plan shall be submitted to Commanding General, US Army Electronics Command, ATTN: AMSEL-PP-EC-4, Fort Monmouth, New Jersey 07703, before implementation of electrical preconditioning.

c. Assemblies failing electrical acceptance tests subsequent to satisfactory completion of electrical preconditioning, may be repaired and electrically retested, accepted and delivered without repeat of electrical preconditioning.

4.7 Service conditions. (Radio only: for antenna test requirements see MIL-R-55656/2.)

4.7.1 Temperature test.

4.7.1.1 Low temperature. The equipment shall be tested in accordance with Method 502, Procedure I, of MIL-STD-810. The storage temperature shall be -65°F for a period of 24 hours and the operating temperature shall be -50°F. Prior to, during, and after testing the equipment shall meet full specification performance for the measurements specified in 4.9.1.

4.7.1.2 High temperature. The equipment shall be tested in accordance with Method 501, Procedure II, of MIL-STD-810. The operating and storage temperatures shall be +155°F. Prior to, during, and after testing the equipment shall meet full specification performance for the measurements specified in 4.9.1.



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4.7.2 Humidity (moisture-resistance) test. The equipment shall be tested in accordance with Method 507, Procedure (II),of MIL-STD-810. Prior to testing,the equipment shall meet full specification performance for the measurements specified in 4.9.1. Measurements shall be made during the periods shown on the figure for the cycle. Do not move equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. If any equipment fails to meet the specified performance during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the equipment still fails to meet the specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test.

4.7.3 Altitude (elevation) test. The equipment shall be tested in accordance with Method 500, Procedure 1, of MIL-STD-810. During and after testing the equipment shall meet full specification performance for the measurements specified in 4.9.2. The chamber may be returned to ambient atmospheric pressure for frequency changes.

4.7.4 Vibration, bounce and shock.

4.7.4.1 Bounce (vibration)test. The equipment shall be tested in accordance with Method 514, Procedure IX, of MIL-STD-810. Prior to and after testing the equipment shall meet full specification performance for the measurements specified in Table VIII. During Part I, jigs shall be provided to observe subassemblies without cases. When practicable, package tester accessories shall be selected from those listed on Drawing SC-A-46439. The adapter plate shall be free of cracks and broken welds, and all unused holes, except the 2-inch breather holes, shall be plugged. No part of the mounted surface shall be less than 7 inches from the edge of the adapter plate. Additional barriers used to safeguard personnel shall not restrain the fore-and aft motion of the adapter plate against the backstop.

4.7.4.2 Shock test: ballistic. The test shall be conducted on the "Shock Testing Machine for Light Weight Equipment" shown in MIL-S-901. The equipment, including shock mounts (if any), shall be secured in its normal operating position to the steel test plate by means of the same fasteners used for vehicular installation of the equipment. The test shall consist of a total of 9 blows; one each 1-foot blow, 3-foot blow, and 5-foot blow on the back, side, and top of the test plate. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissible. Prior to and after testing the equipment shall meet full specification performance for the measurements specified in Table 3.8.



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4.7.4.3 Shock test; bench-handling. The equipment shall be tested in accordance with Method 517, Procedure V, of MIL-STD-810. Prior to and after testing the equipment shall meet full specification performance for the measurements specified in Table VIII.

4.7.5 Drip-proof. The equipment shall be tested in accordance with 4.3(b), Drip Test, of MIL-STD-108. Drops of liquid and/or solid particles shall be falling at an angle of 15° from the vertical when the equipment is mounted in its normal operating position. While the water is being applied the equipment shall be tested in accordance with the requirements of 4.15 and 4.14, power output and noise quieting shall be monitored. There shall be no degradation of performance. Immediately following the drip test, the equipment shall meet full specified performance when tested in accordance with 4.9.1.

4.7.6 Salt fog test. The equipment (including accessories), in its normal operating position, shall be tested in accordance with Method 509, Procedure I, of MIL-STD-810. Operation of the equipment shall not be required and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as any visible degradation of equipment surfaces that can be attributed to flaky, pitted, blistered or otherwise loosened finish or metal surface.

4.7.7 Dust test. The equipment shall be tested in accordance with Method 510, Procedure I, of MIL-STD-810. Prior to and after testing the equipment shall meet full specification performance for the measurement specified in 4.10.

4.7.8 Fungus test. The equipment (including accessories) shall be tested in accordance with Method 508, Procedure I, of MIL-STD-810. Spraying shall include parts and materials on and under the chassis and the interior of the equipment case. The equipment shall remain in an open condition during the test. Operation is not required.

4.8 Standard inspection conditions. Unless otherwise specified, all tests shall be performed at 115 volts ± 5 percent from a single-phase source operating between 47.5 and 63.0 HZ. Tests shall be conducted at room ambient temperature, pressure, and humidity unless otherwise specified.

4.9 Test specifications (service conditions). The equipment shall be tested in accordance with the following test specifications and meet the requirements of this specification. The tests shall be performed during periods as specified in MIL-STD-810.



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4.9.1 Temperature, humidity, and drip. 4.2 and 4.5 of MIL-R-55656/4, except the test be performed at channels 40, 100, 160, 220, 310 and 400 instead of those specified. Tests at line frequencies of 47.5 HZ and 420 HZ and high and low line voltage shall be performed at any one channel. 4.1 of MIL-R-55656/5 with the exception that tests be performed at channels 40, 100, 160, 220, 300 and 400 instead of those specified. 4.6 of MIL-R-55656/4 with the exception that it be performed at channels 40, 160, 280 and 400 instead of those specified. 4.5 of MIL-R-55656/3, except tests be performed at channels 50, 180 and 400 instead of those specified. 4.8 and 4.9 of MIL-R-55656/3, except test be performed at channel 240 instead of those specified. 4.2.7 of MIL-R-55656/3. 4.2 of MIL-R-55656/5, except frequencies below 10 KHZ be deleted from frequency response measurement. 4.5 of MIL-R-55656/5. 4.6.8 of MIL-R-55656/5.

4.9.2 Altitude. 4.2 of MIL-R-55656/4, except that the test be performed at channels 40 and 340 instead of those specified. 4.5 of MIL-R-55656/4, except the test be performed at channel 340 instead of the channel specified. 4.6 of MIL-R-55656/4, except that the test be performed at channel 340 instead of the channel specified. 4.5 of MIL-R-55656/3, except that the test be performed at channel 240 instead of the channel specified. 4.5 of MIL-R-55656/5.

4.10 Operational test service conditions. With the equipment set up in a system, an operational test shall be performed. The test shall consist of voice communications, order wire ringing through the systems plus a visual check of the metering facilities for readings specified in 4.1f and 4.2.1a(f) of MIL-R-55656/5.

4.11 A.C. power line, voltage and frequency tests. The following tests shall be performed at room ambient and at channels 40, 180 and 400. The tests shall be performed at 105 volts 47 HZ, 115 volts 47HZ, 125 volts 47 HZ, 105 volts 420 HZ, 115 volts 420 HZ and 125 volts 420 HZ.



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| INSPECTION | REQUIREMENT PARA |
|---|---|
| 4.2 and 4.5 of MIL-R-55656/4 | 3.13.4.1 |
| 4.5 of MIL-R-55656/3 | 3.13.4.4 |
| 4.2.5 of MIL-R-55656/6 | 3.13.5.1<br>3.13.16 |
| 4.2.2 of MIL-R-55656/5 | 3.13.13 |
| 4.10 of this specification | 3.13.26 |

4.12 Reliability inspection. The accept/reject criteria shall be as specified in Test Plan IV of MIL-STD-781.

4.12.1 Test conditions. The test conditions are as follows:

a. Temperature 40 ± 5°C (95°F to 113°F)

b. Temperature cycling - None

c. Vibration 2.2g's ± 10% peak acceleration value at any non-resonant frequency between 20 and 60 cps measured at the mounting points on the equipment. One eight (8) hour period each week, each equipment shall be vibrated 2 minutes of each operating hour.

d. Equipment on-off cycling - equipment operating cycling shall be on between 48 and 72 hours, off 1 hour.

e. Input voltage - 115 volts, 60 HZ ± 5%.

4.12.2 Reliability test procedure. The reliability test shall be conducted per MIL-R-55656/1 Reliability Test Procedure. During the reliability test, ten (10) equipments shall be operating to accumulate the time to reach, accept or reject decision. Each frequency shall be utilized during the test with frequency changes at least 50 channels apart, and each equipment shall utilize at least 125 channels. Every three days the tuning heads of the equipment under test shall be interchanged.

4.13 Mast and antenna test procedure. The mast and antenna shall be tested in accordance with MIL-R-55656/2 and meet the requirements of 3.14.



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4.14 Radio Set AN/GRC-103( )/(V)1 receiver section test procedure. The Receiver Section of Radio Set AN/GRC-103( )(V)1 shall be tested in accordance with MIL-R-55656/3 and meet the requirements of 3.13.5.

4.15 Radio Set AN/GRC-103( )(V)1 transmitter section test procedure. The Transmitter Section of Radio Set AN/GRC-103( )(V)1 shall be tested in accordance with MIL-R-55656/4 and meet the requirements of 3.13.4.

4.16 Receiver-Transmitter, Order Wire RT-773( )/GRC-103(V) test procedure. The Receiver-Transmitter, Order Wire RT-773( )/GRC-103(V) shall be tested in accordance with MIL-R-55656/14 and meet the requirements of 3.13.26.

4.17 Radio Set AN/GRC-103( )(V)1 terminal test procedure. The Radio Set AN/GRC-103( )(V)1 terminal shall be tested in accordance with MIL-R-55656/5 and meet the requirements of Section 3.

4.18 Component test procedures. The assemblies, subassemblies and modules of Radio Set AN/GRC-103( )(V) shall be tested in accordance with the following specifications:

4.18.1 Receiver Radio R-1329(P)( )/GRC-103(V). The Receiver Radio R-1329(P)( )/GRC-103(V) shall be tested in accordance with MIL-R-55656/6.

4.18.2 Amplifier-Converter, AM-4316/GRC-103(V). The Amplifier Converter AM-4316( )/GRC-103(V) shall be tested in accordance with MIL-R-55656/7.

4.18.3 Transmitter, Radio T-983(P)( )/GRC-103(V). The Transmitter Radio T-983(P)( )/GRC-103(V) shall be tested in accordance with MIL-R-55656/8.

4.18.4 Amplifier Frequency Multiplier AM-4320/GRC-103(V). The Amplifier Frequency Multiplier AM-4320( )/GRC-103(V) shall be tested in accordance with MIL-R-55656/11.

4.18.5 Synthesizer (IRE1A2/5TR1A2). The synthesizer shall be tested in accordance with MIL-R-55656/15.

4.18.6 Restorer, pulse form (IRE1A3). The restorer, pulse form shall be tested in accordance with MIL-R-55656/16.

4.18.7 Limiter-descriminator (1RE1A4). The limiter-discriminator shall be tested in accordance with MIL-R-55656/17.



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4.18.8 Amplifier-frequency multiplier (1RE1A5). The amplifier-frequency multiplier shall be tested in accordance with MIL-R-55656/18.

4.18.9 Amplifier video (1RE1ARI). The amplifier video shall be tested in accordance with MIL-R-55656/19.

4.18.10 Amplifier IF (1RE1AR2). The amplifier IF shall be tested in accordance with MIL-R-55656/20.

4.18.11 Power supply, receiver (1RE1PS1). The power supply, receiver shall be tested in accordance with MIL-R-55656/21.

4.18.12 Synthesizer boards (A3, A5, A6, A7, A8, A9). The synthesizer boards shall be tested in accordance with MIL-R-55656/22.

4.18.13 Modulator-Oscillator (1RE1A2A2/5TR1A2A2). The modulator-oscillator shall be tested in accordance with MIL-T-55656/23.

4.18.14 Frequency divider (1RE1A2A4/5TR1A2A4). The frequency divider shall be tested in accordance with MIL-R-55656/24.

4.18.15 Oscillator, R.F. (1RE1A2Y1/5TR1A2Y1). The oscillator,R.F. shall be tested in accordance with MIL-R-55656/25.

4.18.16 Amplifier voltage regulator (1RE1PS1AR1). The amplifier voltage regulator shall be tested in accordance with MIL-R-55656/26.

4.18.17 Control, alarm (1RE1A1A/5TR1A1A2). The control alarm shall be tested in accordance with MIL-R-55656/27.

4.18.18 Card assembly (1RE1A1A3). The card assembly shall be tested in accordance with MIL-R-55656/28.

4.18.19 Tuner, radio frequency (2A1). The tuner, radio frequency shall be tested in accordance with MIL-R-55656/29.

4.18.20 Control indicator (2A2). The control indicator shall be tested in accordance with MIL-R-55656/30.

4.18.21 Dummy load (2A3). The dummy load shall be tested in accordance with MIL-R-55656/31.

4.18.22 Electronic switch (2A4). The electronic switch shall be tested in accordance with MIL-R-55656/32.



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4.18.23 Power supply (2PS1). The power supply shall be tested in accordance with MIL-R-55656/33.

4.18.24 Duplexer (2A1A1). The duplexer shall be tested in accordance withMIL-R-55656/34.

4.18.25 Converter frequency electronic (2A1A2). The converter frequency electronic shall be tested in accordance with MIL-R-55656/35.

4.18.26 Monitor, power (2A1A5). The monitor, power shall be tested in accordance with MIL-R-55656/36.

4.18.27 Amplifier, RF (2A1AR1). The amplifier, RF shall be tested in accordance with MIL-R-55656/37.

4.18.28 Filter assembly band pass (2A1A1A1). The filter assembly band pass shall be tested in accordance with MIL-R-55656/38.

4.18.29 Filter, low pass (2A1A1FL1). The filter, low pass shall be tested in accordance with MIL-R-55656/39.

4.18.30 Network, impedance match (2A1A1Z1). The network, impedance match shall be tested in accordance with MIL-R-55656/40.

4.18.31 Frequency multiplier (2A1A2A1). The frequency multiplier shall be tested in accordance with MIL-R-55656/41.

4.18.32 Mixer (2A1A2A2). The mixer shall be tested in accordance with MIL-R-55656/42.

4.18.33 Filter assembly (2A1A2A3). The filter assembly shall be tested in accordance with MIL-R-55656/43.

4.18.34 Amplifier PRE IF (2A1A2A5). The amplifier PRE IF shall be tested in accordance with MIL-R-55656/44.

4.18.35 Cavity, tuned (2A1A1AZ1). The cavity, tuned shall be tested in accordance with MIL-R-55656/45.

4.18.36 Cavity, tuned (2A1A1A1Z2). The cavity, tuned shall be tested in accordance with MIL-R-55656/46.

4.18.37 Cavity, tuned (2A1A1A1Z3). The cavity, tuned shall be tested in accordance with MIL-R-55656/47.



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4.18.38 Cavity, tuned (2A1A1A1Z4). The cavity, tuned shall be tested in accordance with MIL-R-55656/48.

4.18.39 Cavity, tuned (2A1A1A1Z5). The cavity, tuned shall be tested in accordance with MIL-R-55656/49.

4.18.40 Control alarm (5TR1A3). The control alarm shall be tested in accordance with MIL-R-55656/50.

4.18.41 Amplifier frequency multiplier (5TR1A4). The amplifier frequency multiplier shall be tested in accordance with MIL-R-55656/51.

4.18.42 Amplifier monitor (5TR1A5). The amplifier monitor shall be tested in accordance with MIL-R-55656/52.

4.18.43 Power supply, transmitter (5TR1PSI). The power supply, transmitter shall be tested in accordance with MIL-R-55656/53.

4.18.44 Amplifier voltage regulator (5TR1PS1AR1). The amplifier voltage regulator shall be tested in accordance with MIL-R-55656/54.

4.18.45 Power supply subassembly (5TR1PSIA2). The power supply subassembly shall be tested in accordance with MIL-R-55656/55.

4.18.46 Fan, centrifugal (5A2B1). The fan, centrifugal shall be tested in accordance with MIL-R-55656/56.

4.18.47 Electronic switch (6A1). The electronic switch shall be tested in accordance with MIL-R-55656/57.

4.18.48 Frequency multiplier assembly (6A2). The frequency multiplier assembly shall be tested in accordance with MIL-R-55656/58.

4.18.49 Control indicator (6A3). The control indicator shall be tested in accordance with MIL-R-55656/59.

4.18.50 Control RF power level (6A5). The control rf power level shall be tested in accordance with MIL-R-55656/60.

4.18.51 Amplifier RF (6AR1). The amplifier RF shall be tested in accordance with MIL-R-55656/61.

4.18.52 Power monitor (6AR1A3). The power monitor shall be tested in accordance with MIL-R-55656/62.



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4.18.53 Circuit card assembly (6AR1A2A2). The circuit card assembly shall be tested in accordance with MIL-R-55656/63.

4.18.54 Filter assembly (6AR1A2A4). The filter assembly shall be tested in accordance with MIL-R-55656/64.

4.18.55 Filter low pass (6AR1A2FL1). The filter low pass shall tested in accordance with MIL-R-55656/65.

4.18.56 Panel distribution (9A1). The panel distribution shall be tested in accordance with MIL-R-55656/66.

4.18.57 Panel monitor(9A2). The panel monitor shall be tested in accordance with MIL-R-55656/67.

4.18.58 Converter, telephone signal (9A3). The converter, telephone signal shall be tested in accordance with MIL-R-55656/68.

4.18.59 Amplifier assembly (9A4). The amplifier assembly shall be tested in accordance with MIL-R-55656/69.

4.18.60 Dummy load (16) The dummy load shall be tested in accordance with MIL-R-55656/70.

4.18.61 Continuity test of subassemblies. The continuity test of subassemblies shall be in accordance with MIL-R-55656/71.

4.18.62 RF interconnecting cable (CG-3444/U). In addition to the requirements of the applicable specifications, the Cable Assembly CG-3444/U shall meet MIL-R-55656/72.

4.18.63 Antenna feed cable (CG-3443/U). In addition to the requirements of the applicable specifications, the Cable Assembly CG-3443/U shall meet MIL-R-55656/73.



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4.19 Visual and mechanical inspection. Parts and equipment shall be examined for the defects listed in MIL-STD-252.

4.20 Welding inspection. Welding shall be inspected for non-visual defects by the magnetic particle method specified in MIL-M-11473 and by the ultrasonic method using a reflectoscope, as made by Sperry Products, Incorporated, Danbury, Connecticut, or equal. Both tests shall be performed on each sample unit.

4.21 Electromagnetic interference tests. The AN/GRC-103( )(V)1 shall be tested for compliance to the electromagnetic interference requirements of 3.13.4.5; 3.13.5.3 and 3.23. All test methods and setups shall be in accordance with the requirements of MIL-STD-462 and appendix of MIL-STD-461. Broadband emission tests shall be performed with the AN/GRC-103( )(V)1 operating at 310 MHZ. All tests except CE06 and CS08 shall be performed with the equipment operating in a full duplex mode.

4.22 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.



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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified.

5.1.1 Level A.

5.1.1.1 Cleaning. Each Radio Set AN/GRC-103(V) shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each Radio Set AN/GRC-103(V) shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. Unit packaging shall be in accordance with the method prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be packaged Method I C-1.

5.1.1.4.2 Transmitter, Radio T-983(P)/GRC-103(V) with Amplifier Frequency Multiplier AM-4320/GRC-103(V). Secured within Case CY-4637/GRC-103(V), Transmitter, Radio T-983(P)/GRC-103(V) with Amplifier Frequency Multiplier AM-4320/GRC-103(V) shall be packaged Method III as follows: Cushion the Case CY-4637/GRC-103(V), with its contents, on all surfaces with cells or pads, or both, fabricated of fiberboard conforming to PPP-F-320, W5c, weather resistant, designed and fabricated to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned item within a close-fitting fiberboard box conforming to PPP-B-636, W5C, weather resistant. Box closure shall be as specified in appendix of the box specification.

5.1.1.4.3 Receiver, Radio R1329(P)/GRC-103(V) with Amplifier Converter AM-4316/GRC-103(V). Secured within Case CY-4638/GRC-103(V), Receiver, Radio R1329(P)/GRC-103(V) with Amplifier Converter AM-4316/GRC-103(V) shall be packaged Method III as specified in 5.1.1.4.2.

5.1.1.4.4 Receiver-Transmitter, Order Wire RT-773/GRC-103(V) with Handset H-60/PT. Secured within Case CY-4635/GRC-103(V), Receiver-Transmitter, Order Wire RT-773/GRC-103(V) with Handset H-60/PT shall be packaged Method III as follows: Place the Handset H-60/PT within the compartment provided in the case and fill all voids with paperboard conforming to PPP-P-291, Type 1. Close cover of compartment and secure. Continue packaging as specified in 5.1.1.4.2.



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5.1.1.4.5 Antenna AS-1852/GRC-103(V). Antenna AS-1852/GRC-103(V) shall be packaged Method III as follows: Remove T-shaped dipole from center of antenna and close antenna to its smallest dimension. Secure by means of tightening knobs on both sides. Cushion the antenna on all surfaces by means of pads or cells, or both, fabricated of fiberboard conforming to PPP-F-320, W5c, weather resistant. Secure the cushioning by means of pressure-sensitive tape conforming to PPP-T-76. Cushion the dipole by wrapping in paperboard conforming to PPP-P-291, Type III. Secure wrapping by means of pressure-sensitive tape conforming to PPP-T-76. Place the cushioned antenna, together with the cushioned dipole, within a close-fitting fiberboard box conforming to PPP-B-636, W5c class, weather resistant. Box closure shall be as specified in the appendix of the box specification.

5.1.1.4.6 Accessories in waterproof bag. Accessories shall be packaged Method III as follows: Place the designated components within the spaces provided in the bag. Close the bag cover and secure with straps. Place the accessory bag, with contents, within a close-fitting fiberboard box conforming to PPP-B-636, W5c, weather resistant. Box closure shall be in accordance with the applicable box specification.

5.1.1.4.7 Cable assemblies. Cable assemblies shall be individually packaged Method III as follows: Coil cables in proportionate dimension. Tie in three places with cotton twine conforming to T-T-871. No further packaging required.

5.1.1.4.8 Adapter, connector. Adapter, connector shall be packaged Method III as follows: Cushion item by wrapping in paperboard conforming to PPP-P-291, type 1. Secure wrapping with pressure-sensitive tape conforming to PPP-T-97. No further packaging required.

5.1.1.4.9 Dummy Load, Electrical DA-437/GRC-103(V). Dummy Load, Electrical DA-437/GRC-103(V) shall be individually packaged Method III as specified in 5.1.1.4.8.

5.1.1.4.10 Consolidated package. Place the items packaged as specified in 5.1.1.4.6 through 5.1.1.4.9 within a close-fitting fiberboard box conforming to PPP-B-636, W5c, class weather resistant. Fill all voids with paperboard conforming to PPP-P-291, type 1, to prevent movement. Box closure shall be as specified in the appendix of the box specification.



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5.1.1.4.11 Mast AB-952/GRC-103(V). Mast AB-952/GRC-103(V) shall be packaged Method III as follows: Each mast assembly shall be collapsed to its smallest dimension and secured within the end plates by means of straps, and cables provided. No further packaging required.

5.1.2 Level C. Radio Set AN/GRC-103(V)1 shall be packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified.

5.2.1 Level A. Components of Radio Set AN/GRC-103(V) , packaged as specified in 5.1.1.4.2 through 5.1.1.4.4, shall be packed within a close-fitting fiberboard box conforming to MIL-B-43291 or PPP-B-640, class weather resistant. Place the technical literature, packaged as specified in 5.1.1.4.1, in the box directly under the lid of the box. Box closure shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Antenna AS-1852/GRC-103(V) and consolidated package. Packaged as specified in 5.1.1.4.5 and 5.1.1.4.10, Antenna AS-1852/GRC-103(V) and consolidated package shall be packed within a close-fitting fiberboard box conforming to MIL-B-43291 or PPP-B-640, class weather resistant. Box closure shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Mast AB-952/GRC-103(V). Packaged as specified in 5.1.1.5.1, Mast AB-952/GRC-103(V) shall be packed in a close-fitting fiberboard box conforming to MIL-B-43291 or PPP-B-640, class weather resistant. Box closure shall be as specified in the appendix of the applicable box specification.

5.2.1.3 Reinforcing. Shipping containers shall be reinforced by pressure-sensitive filament tape bonding or non-metallic strapping conforming to PPP-T-97, type IV and PPP-S-760, type II, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.



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5.2.2 Level B. Radio Set AN/GRC-103(V) , packaged as specified in 5.1, shall be packed as specified in 5.2.1.

5.2.3 Level C. Radio Set AN/GRC-103(V) , packaged as specified in 5.1, shall be packed as specified in 5.2.1 except that the fiberboard boxes shall be of the type, size and kind commonly used for the purpose, and shall comply with the rules and regulation of the common carrier as applicable to the mode of transportation. Packing shall be accomplished in a manner that will afford adequate protection to the package and its contents during shipment from the supply source to the first receiving activity.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Box sequence. Box sequence shall be as follows:

| | Box No. |
|---|---|
| Components of Radio Set AN/GRC-103(V) packaged as specified in 5.1.1.4.2 through 5.1.1.4.4 | 1/3 |
| Antenna AS-1852/GRC-103(V) and intermediate package | 2/3 |
| Mast AB-952/GRC-103(V) | 3/3 |

6. NOTES

This section is not applicable to this specification

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APPENDIX I

DATED 15 February 1969

10. SCOPE

10.1 This appendix covers an Index of applicable test specifications.

20. INDEX

| MIL NO. | TEST SPECIFICATION |
|---|---|
| MIL-R-55656/1 | Reliability Test Procedure |
| MIL-R-55656/2 | Mast and Antenna Test Procedure |
| MIL-R-55656/3 | Radio Set AN/GRC-103(V)1 Receiver Section |
| MIL-R-55656/4 | Radio Set AN/GRC-103(V)1 Transmitter Section |
| MIL-R-55656/5 | Radio Set AN/GRC-103(V)1 Terminal |
| MIL-R-55656/6 | Receiver Radio, R-1329(P)/GRC-103(V) |
| MIL-R-55656/7 | Amplifier, Converter AM-4316/GRC-103(V) |
| MIL-R-55656/10 | Transmitter, Radio T-983(P)/GRC-103(V) |
| MIL-R-55656/11 | Amplifier, Frequency Multiplier AM-4320/GRC-103(V) |
| MIL-R-55656/14 | Receiver-Transmitter, Order Wire RT-773/GRC-103(V) |
| MIL-R-55656/15 | Synthesizer 1RE1A2/5TR1A2 |
| MIL-R-55656/16 | Restorer, Pulse Forms 1RE1A3 |
| MIL-R-55656/17 | Limiter-Discriminator 1RE1A4 |
| MIL-R-55656/18 | Amplifier-Frequency Multiplier 1RE1A5 |



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| MIL-NO. | TEST SPECIFICATION |
|---|---|
| MIL-R-55656/19 | Amplifier Video 1RE1AR1 |
| MIL-R-55656/20 | Amplifier IF 1RE1AR2 |
| MIL-R-55656/21 | Power Supply, Receiver 1RE1PS1 |
| MIL-R-55656/22 | Synthesizer Boards A3, A5, A6, A7, A8, A9 |
| MIL-R-55656/23 | Modulator - Oscillator 1RE1A2A2/ 5TR1A2A2 |
| MIL-R-55656/24 | Frequency Divider 1RE1A2A4/ 5TR1A2A4 |
| MIL-R-55656/25 | Oscillator, R. F. 1RE1A2Y1/ 5TR1A2Y1 |
| MIL-R-55656/26 | Amplifier Voltage Regulator 1RE1PS1AR1 |
| MIL-R-55656/27 | Control, Alarm 1RE1A1A/5TR1A1A2 |
| MIL-R-55656/28 | Card Assembly 1RE1A1A3 |
| MIL-R-55656/29 | Tuner, Radio Frequency 2A1 |
| MIL-R-55656/30 | Control Indicator 2A2 |
| MIL-R-55656/31 | Dummy Load 2A3 |
| MIL-R-55656/32 | Electronic Switch 2A4 |
| MIL-R-55656/33 | Power Supply 2PS1 |
| MIL-R-55656/34 | Duplexer 2A1A1 |
| MIL-R-55656/35 | Converter Frequency Electronic 2A1A2 |
| MIL-R-55656/36 | Monitor, Power 2A1A5 |
| MIL-R-55656/37 | Amplifier, RF 2A1AR1 |



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| MIL-NO | TEST SPECIFICATION |
|---|---|
| MIL-R-55656/38 | Filter Assembly Band Pass 2A1A1A1 |
| MIL-R-55656/39 | Filter, Low Pass 2A1A1FL1 |
| MIL-R-55656/40 | Network, Impedance Match 2A1A1Z1 |
| MIL-R-55656/41 | Frequency Multiplier 2A1A2A1 |
| MIL-R-55656/42 | Mixer 2A1A2A2 |
| MIL-R-55656/43 | Filter Assembly 2A1A2A3 |
| MIL-R-55656/44 | Amplifier PRE IF 2A1A2A5 |
| MIL-R-55656/45 | Cavity, Tuned 2A1A1AZ1 |
| MIL-R-55656/46 | Cavity, Tuned 2A1A1A1Z2 |
| MIL-R-55656/47 | Cavity, Tuned 2A1A1A1Z3 |
| MIL-R-55656/48 | Cavity, Tuned 2A1A1A1Z4 |
| MIL-R-55656/49 | Cavity, Tuned 2A1A1A1Z5 |
| MIL-R-55656/50 | Control Alarm 5TR1A3 |
| MIL-R-55656/51 | Amplifier Frequency Multiplier 5TR1A4 |
| MIL-R-55656/52 | Amplifier Monitor 5TR1A5 |
| MIL-R-55656/53 | Power Supply, Transmitter 5TR1PS1 |
| MIL-R-55656/54 | Amplifier Voltage Regulator 5TR1PS1AR1 |
| MIL-R-55656/55 | Power Supply Subassembly 5TR1PS1A2 |
| MIL-R-55656/56 | Fan, Centrifugal 5A2B1 |
| MIL-R-55656/57 | Electronic Switch 6A1 |



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| MIL NO. | TEST SPECIFICATION |
|---|---|
| MIL-R-55656/58 | Frequency Multiplier Assembly 6A2 |
| MIL-R-55656/59 | Control Indicator 6A3 |
| MIL-R-55656/60 | Control RF Power Level 6A5 |
| MIL-R-55656/61 | Amplifier RF 6AR1 |
| MIL-R-55656/62 | Power Monitor 6AR1A3 |
| MIL-R-55656/63 | Circuit Card Assembly 6AR1A2A2 |
| MIL-R-55656/64 | Filter Assembly 6AR1A2A4 |
| MIL-R-55656/65 | Filter Low Pass 6AR1A2FL1 |
| MIL-R-55656/66 | Panel Distribution 9A1 |
| MIL-R-55656/67 | Panel Monitor 9A2 |
| MIL-R-55656/68 | Converter, Telephone Signal 9A3 |
| MIL-R-55656/69 | Amplifier Assembly 9A4 |
| MIL-R-55656/70 | Dummy Load 16 |
| MIL-R-55656/71 | Continuity Tests |
| MIL-R-55656/72 | Cable Assembly, RF, CG-3444( )/U |
| MIL-R-55656/73 | Cable Assembly, RF, CG-3443( )/U |



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APPENDIX II

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10. SCOPE

10.1 This appendix covers descriptions of the special test equipment and accessories.

20. TEST METHODS AND PROCEDURES

20.1 The contractor shall apply the test methods and procedures described by the applicable test specifications listed in Appendix I as specified. Where a test jig is not described herein, the contractor shall fabricate a suitable jig subject to approval of the contracting officer's representative.

30. SPECIAL TEST EQUIPMENT AND ACCESSORIES

30.1 FM/AM Modulation Meter (Marconi Instruments TF2300). The Marconi Instruments' FM/AM Modulation Meter type TF2300 is essentially a low-sensitivity superheterodyne receiver having very linear demodulators for both frequency modulated and amplitude modulated signals. It accepts carrier frequencies from 4 MHz up to 1 GHz with a sensitivity varying from 20 mV below 250 MHz down to 100 mV above 500 MHz. Up to 3 volts rms may be applied to the 50 ohm input. Modulating frequencies up to 150 kHz can be used. The instrument is equipped with an output meter directly calibrated in both kHz peak deviation and percentage modulation for FM and AM respectively. Five ranges of peak deviation from 0-5 kHz up to 0-500 kHz are provided for FM and two ranges of 0-30% and 0-100% for AM. The full scale accuracy is ± 5%. A high impedance i-f output is provided in addition to the standard 600 ohm output. There are also facilities for the insertion of up to three switch selected crystals to lock the internal oscillator, or for the alternative connection of an external oscillator. Two standard de-emphasis filters and a 15 kHz low-pass filter can also be switched into the output amplifier as required. The equipment is all solid-state and may be powered from supply mains or external battery.

30.2 Level Meter (Siemens 3D335d3a). The Siemens' selective level meter type 3D335d3a is a heterodyne type detector specially designed for transmission use, with four frequency translations and a final i-f of 3 kHz. It accepts frequencies from 10 kHz up to 17 MHz either in 100 kHz spectrum steps or continuously tuned over the whole range. Fine tuning between -10 kHz and +100 kHz, with respect to the main tuning setting, is provided and readability is approximately 200 Hz. Accuracy with spectrum step tuning amounts to $\pm 2 \times 10^{-5} \pm 300$ Hz. The input impedance is adjustable in five steps from 65 ohm to 150 ohm, and may be terminated or high impedance for each setting. Meter is adjusted to read dBm directly for any input impedance setting. The input attenuator provides for input levels from +30 dBm to -70 dBm and an additional attenuator may be switched in to extend ragne to -90 dBm. Accuracy is of the order of ± 0.2 dB. Selectivity is adjustable to 200 Hz or 2 kHz bandwidth. The equipment is normally powered from supply mains.

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30.3 Variable Frequency Power Supply (Empire Devices VP-410). The Empire Devices' variable frequency power supply is essentially a low frequency oscillator followed by a high power amplifier. It covers a frequency range of 40 Hz to 600 Hz, or 400 Hz to 6 kHz with X10 switching. Output is nominally 115 volts rms at 500 watts and may be varied within the range 75V ac to 150V ac indicated on the output voltmeter. The equipment is powered from supply mains.

30.4 Attenuator, Precision, 0-16 dB, 75 OHM (Ortho Filter 36A). The Ortho Filter precision attenuator model 36A is a completely screened rack-mounting component. It has two step switch controls, one covering 0 to 15 dB in 1 dB steps and the second 0 to 1.5 dB in 0.1 dB steps, permitting the selection of any attenuation from 0 to 16.5 dB in 0.1 dB increments. Impedance is maintained at 75 ohm at all settings and input and output connections are BNC at the rear of the panel.

30.5 Modem Test Set, 30 MHz. The 30 MHz modem test set (T-43363) consists of an oscillator-modulator and a demodulator. The 30 MHz oscillator has an accuracy of ±10 kHz and is of the collector voltage modulating type using carrier frequency stabilization in a closed loop auto frequency control circuit. The output power is -6 dBm ±1 dB into 50 ohms. A fraction of this is fed to a limiter-discriminator circuit, followed by a dc amplifier, to provide the auto frequency control voltage for the oscillator. Internal modulation is from a 1200 Hz ± 2% oscillator and peak frequency deviations of ±300 kHz ± 2%, ±180 kHz ± 2%, ±75 kHz ± 3% and ±45 kHz ± 3% are obtainable by switching. External modulation may also be applied in the range 50 Hz to 1 MHz with a response flat within ±0.5 dB. The input impedance is 75 ohm and the modulation sensitivity 2.8 kHz/mV. The demodulator input signal at 30 MHz ±30 kHz and at a level between -40 dBm and -10dBm into 50 ohms is fed to an auto gain controlled i-f amplifier through a 30 MHz band pass filter. The amplified signal is demodulated in a limiter-discriminator followed by a low pass filter. The response over the range 50 Hz to 1 MHz is flat within ±0.3 dB and the output impedance is 50 ohm, with a sensitivity of 0.43 mV/kHz. A regulated dc supply of 24 volts with negative ground is required and current drain is 240 mA.



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30.6 PCM Test Set T-43922. The PCM test set T-43922 consists of a pulse generator unit and shift register and comparator unit. The pulse generator produces an eight-bit repetitive pulse train of the sequence 10111000. This pulse train is normally fed into the AN/GRC-103 system under test, transmitted, received, and regenerated. It is then fed into the shift register where it is subjected to an eight-bit delay. The comparator then compares the output of the shift register with the input. If there are no errors, both waveforms will be identical and there will be no error signal. If, however, random noise appears on the system, the pulse trains will no longer be identical, error signals will result and may be detected by an electronic counter. A recorder may be used with the counter if required. The equipment is powered from supply mains.

30.7 Random PCM Generator T-45838. The PCM test set T-45838 consists of a 576 kHz crystal oscillator followed by a frequency divider which produces a 288 kHz square wave. This wave train passes through a gate controlled by a random noise generator and hence produces a random pulse output. A dc amplifier has a switched input so that either the full waveform or the random output may be selected. Separate provision is also made for the insertion of a 25% order wire signal. Both 91 ohm and 75 ohm outputs are available from the amplifier at levels of 2 volts peak to peak or 130 millivolts respectively. Output may also be taken via a low pass filter with gaussian characteristics and a phase delay network (40 degrees at 200 Hz). The equipment requires a -12/0/+12 volt dc supply.

30.8 Dummy Load, Electrical (DA 437/GRC-103(V)). The dummy load, electrical used in many of the tests, is an accessory supplied with the AN/GRC-103(V)1 radio set. It consists of a 50 ohm load resistor with cooling fins, capable of dissipating 50 watts continuously, and having a VSWR over the frequency range 220 MHz to 1 GHz not exceeding 1.5:1.

30.9 RF Amplifier Test Jig T-45884. The RF amplifier test jig T-45884 consists of a standard amplifier, radio frequency 2A1AR1 as used in the AN/GRC-103(V)1, mounted in a jig with suitable input and output connections. The amplifier has an input and output impedance of 50 ohms nominal and a minimum bandwidth of 185 MHz (220 MHz to 405 MHz). The maximum input level is -10 dBm and the gain 27 dB ±3 dB. The input VSWR is 2.0:1 maximum, the noise figure not greater than 6 dB and the amplifier is flat within 0.05 dB over ±0.5 MHz or within 4 dB over ±92.5 MHz. A -12/0/+12 volt dc supply is required.



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30.10 Amplifier, Video Test Jig T-45064. The video amplifier test jig T-45064 provides connection and indicating facilities, similar to those normally available on the complete equipment, for the video amplifier unit under test. It is provided with a panel meter switched to read FDM or 12 CH PCM, for setting up output levels, together with warning lamps for PCM OUT, LOW SIGNAL ALARM, SQUELCH FDM. Video and PCM outputs are brought out to suitable connectors for test purposes. A -12/0/+12 volt dc supply is required.

30.11 Order Wire Test Jig T-41139. The order wire test jig T-41139 provides connection and indicating facilities, similar to those normally available on the complete equipment, for the receiver/transmitter signal amplifier and the order wire amplifier boards from the order wire unit. It provides suitable input and output connectors and calling switches and indicator lamp. A 12 volt dc supply is required.

30.12 Order Wire System Test Jig T-46686. The order wire system test jig T-46686 provides for connections to the microphone and receiver terminals in the two order wire handsets of a one way system test set up for sidetone, frequency response and net loss tests. It provides for an audio signal input from an oscillator, via a suitable R-C circuit to the microphone terminals of a handset under test. It also provides a terminating resistor for vacuum tube voltmeter readings across the handset receiver terminals, together with connections for the voltmeter to input or output as required. No power supply is required for this jig.

30.13 Coaxial Switch T-43269. The coaxial switch T-43269 is a screened coaxial switch of 75 ohm impedance, generally similar to the Telonic TS-100A.

30.14 Return Loss Bridge T-45072. The return loss bridge T-45072 is a Wheatstone bridge with fixed arms of 600 ohms and coaxial connections for a standard termination and the unknown. An unbalanced to balanced transformer is incorporated to feed the bridge from the coaxial generator input. A calibrating capacitor is also incorporated across the standard termination connector, and a coaxial connection provided for the vacuum tube voltmeter indicator.

30.15 Frequency Changer T-43378. The frequency changer T-43378, used for loop test set ups is a coaxial diode holder with N-type male connector on one end, and a BNC female connector on the other. The diode used is 1N21B and the BNC connector is shunted by a 5 uuF capacitor.



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30.16 Matching Pads - 91/75 OHM T-41699 - 50/75 OHM T-41374. These two matching pads are simple screened L-section pads. T-41699 matches 91 ohms to 75 ohms with a loss of 3.9 dB. T-41374 matches 75 ohms to 50 ohms with a loss of 5.72 dB. Resistors used have a ± 1% tolerance and all connectors are BNC female.

30.17 Terminations - 91 OHM T-41369 - 75 OHM T-41133 - 600 OHM T-41666. These three terminations have BNC female connectors shunted by resistors of appropriate value with a tolerance of 1%. The 600 ohm termination is also provided with banana type plugs for direct connection to a vacuum tube voltmeter when required.

30.18 Cable Assemblies. The following cable assemblies are used in the various test set ups called up in Part I of this specification:

| DESCRIPTION | LENGTH | TYPE |
|---|---|---|
| a. Cable, 50 ohm, RG-214A/U with UG-573B/U connectors (C type) | 1 Ft 6 In<br>3 Ft 6 In | |
| b. Cable, 93 ohm, RG-62A/U, with UG-260D/U connectors (BNC) | 4 Ft -<br>1 Ft - | T-47204<br>T-47205 |
| c. Cable, 75 ohm, RG-59A/U, with UG-260D/U connectors (BNC) | 8 Ft -<br>6 Ft -<br>4 Ft -<br>2 Ft -<br>1 Ft - | T-47206<br>T-47207<br>T-47208<br>T-47209<br>T-47210 |
| d. Cable, 50 ohm, RG-58C/U, with UG-88/U connectors (BNC) | 4 Ft -<br>3 Ft -<br>1 Ft - | T-47211<br>T-47212<br>T-47213 |
| e. Cable, 50 ohm, RG-8A/U, with UG-21B/U connectors (N type) | 3 Ft - | T-47214 |
| f. Cable Assy., Special, for Order Wire Tests | - - | T-47215 |



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30.19 Adapters. The following standard adapters are used:

| DESCRIPTION | TYPE |
|---|---|
| a. N female to N female | UG-29B/U |
| b. N male to N male | UG-57B/U |
| c. N male to BNC female | UG-201/U |
| d. N female to BNC male | UG-349A/U |
| e. N male to C female | UG-564/U |
| f. N female to C male | UG-565A/U |
| g. C female to C female | UG-643/U |
| h. BNC female to BNC female | UG-914/U |
| j. N male to BNC male | UG-1034/U |
| k. 'T' - N male/N female/N female | UG-464/U |
| m. 'T' - N female/N male/N female | UG-107B/U |
| n. 'T' - BNC female/BNC male/BNC female | UG-274/U |
| p. 'T' - BNC female/BNC female/BNC female | DIC-2005/5981 |
| q. 'T' - BNC male/BNC male/BNC male | 5378 |



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APPENDIX III

DATED 15 February 1969

10. SCOPE

10.1 This appendix covers the engineering changes that are not reflected in the model and shall be incorporated in the Radio Set AN/GRC-103( )(V) equipments produced under this specification. Changes shall be accomplished without impairing performance. Modifications are the contractor's responsiblility and shall be submitted to the CODR for approval prior to incorporation into the equipment. The government shall assume no responsibility for the accuracy of any supplied information.

20. ENGINEERING CHANGES

20.1 This section describes the design characteristics of the model which are subject to change. Any difference between the model characteristics and the requirements of section 3 represents the required engineering change.

20.2 Drip-proof. The model has not been designed or tested for this requirement (see 4.7.5).

20.3 Salt fog. The model has not been designed or tested for this requirement (see 4.7.6).

20.4 Dust. The model has not been designed or tested for this requirement (see 4.7.7).

20.5 Fungus. The model has not been designed or tested for this requirement (see 4.7.8).

20.6 Electromagnetic interference characteristics. The equipment shall conform to the requirements of MIL-E-55301 for tactical equipment (see also Table A-1 of MIL-STD-416A). When tested for cable and case leakage and antenna conducted interference, the performance of the model shall govern (see 4.21).



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20.7 Low temperature test. The equipment shall be tested in accordance with Method 502, Procedure I of MIL-STD-810. The storage temperature shall be -65°F for a period of 24 hours, and the operating temperature shall be -35°F. Prior to, during, and after testing, the equipment shall meet full specification performance for the measurements specified in 4.9.1 (see 4.7.1.1 ).

20.8 High temperature test. The equipment shall be tested in accordance with Method 501, Procedure II of MIL-STD-810. The storage temperature shall be +155°F and the operating temperature shall be +125°F. Prior to, during, and after testing, the equipment shall meet full specification performance for the measurements specified in 4.9.1 (see 4.7.1.2 ).

20.9 Altitude (elevation) test. The equipment shall be tested in accordance with Method 501, Procedure I of MIL-STD-810, except change pressure in step 2 to simulate 25,000 feet. During and after testing, the equipment shall meet the full specification performance for the measurements specified in 4.9.2. The chamber may be returned to ambient barometric pressure for frequency changes during operating tests of step 3 (see 4.7.3 ).

30. EXCEPTIONS

30.1 This section describes characteristics of the model which are not desirable and not to be produced without specific consent of the CODR.

30.2 Standard parts. An unknown number of non-standard parts are in the model. Their use does not constitute a waiver from the requirements of this specification or MIL-P-11268, nor an automatic approval for their use in the equipment produced.

30.3 Mast height. The mast was designed to meet the 50 foot height but has been tested only to 35 feet.



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# SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-R-55656(EL) RADIO SET AN/GRC-103( )(V) (BASIC)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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